

Block

# 4

## LAYOUT AND DESIGNING

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## BLOCK INTRODUCTION: LAYOUT & DESIGNING

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This is the fourth block of **Course MJM-022: Layout & Designing**. There are four units in this block that covers all aspects and dimensions of layout and design of a newspaper or magazine. With the advent of advanced technological features in the digital world, the print media is forced to present its news content in an attractive manner using various innovative design techniques and multiple media elements. Basically this has to be done in order to attract the readers to the print media and more particularly draw the reader's attention towards each and every news content. All the four units deal with the latest design challenges faced by the print media.

**Unit 12: Page Layout** describes the basic nuances of layout of a newspaper or a magazine. It begins with giving you specific information about how a layout is being developed in a newsroom - such as stages of layout, various elements of layout, important characteristics and good features of a layout and finally gives a detailed account on various layout formats for diverse print media platforms.

**Unit 13: Principles of Page Design** is an extension of earlier Unit. Here you will learn in-depth details on design principles and its relevance. Basically you will learn the basic concepts of design, how a design idea is being developed and processed, elements of design, and more importantly you will learn the principles of design. Even though design is a creative process, you there are a number of principles for a good design. The ultimate aim of any principles of design is to maintain the white space in an economical way by presenting appealing designs.

**Unit 14: Typography and Image** gives you an overview of characteristics and features of various types of fonts and different kinds of image formats that are being used in the print media. In the typographic segment, you will learn the evolution of typography, principles and elements of typography, and role of typography in design. The role of typography in the digital world has also been covered. In the image segment, you will learn about the basic formats of various images, the required image format for print as well as digital news platforms. A basic outline is being given on editing and optimization of image for the digital world.

**Unit 15: Infographics** is a new discipline in the print media. This Unit takes you through various types of infographics, evolution of the visual presentation of news content, different elements of infographics etc. which are being covered extensively in this Unit. Infographics is a new feature of print media adopted to face the challenges of digital media. Here in the infographics, the news content is presented in a gist format using various media elements in a colourful way.

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## UNIT 12 PAGE LAYOUT

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### Structure

- 12.0 Introduction
- 12.1 Learning Outcomes
- 12.2 Page Layout- the Definition
- 12.3 Stages of Layout
- 12.4 Elements of Layout
- 12.5 Importance of a Good Layout
- 12.6 Layout Formats for Various Print Publications
- 12.7 Let Us Sum Up
- 12.8 Keywords
- 12.9 Further Readings
- 12.10 Check Your Progress: Possible Answers

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### 12.0 INTRODUCTION

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Everything in the world that we see has a fixed pattern as per which we perceive them. The placement of text along with the various other elements arranged on a page is called as the layout.

There is nothing in the world that can be created in a vacuum. Every plan, be it of a building or any new project needs a strong foundation. Similarly, like a building, a plan is needed for a graphic design. A graphic design plan is called a layout.

Page layout is the part of graphic design that deals in the arrangement and style treatment of elements (contents) on a page. A page of a book or a magazine is similar to the brick of a building. Every page has a definite layout. When it comes to the layout, there are several elements within it that need to be taken care of.

Let us now proceed to understand the same in detail.

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### 12.1 LEARNING OUTCOMES

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After going through the contents of this Unit, you will be able to:

- describe the basics of layout as a process.
- use the concept of page layout in your day –to- day life and above all for educational purposes.
- apply page layout in the teaching-learning process also.

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### 12.2 PAGE LAYOUT- THE DEFINITION

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Design is the creation of a plan for the construction of an object or a system. It helps in laying the foundation in the formation of any new object or system. And, when it comes to the page designing, layout plays a pivotal role in graphic design.

Layout refers to the arrangement of various elements on a page. It usually refers to a specific placement of image, text and style. Understanding the layout of design is very important because of the fact that it is the layout which makes a page presentable, readable and appealing to the eyes. If the layout is not correctly understood, there is a probability that the message that needs to be conveyed might be lost and the cost as well as the labour invested in putting together the content would go in vain.

Irrespective of how good the content is, if the arrangement of the content is not organized, the impact of the same would be nil. Effective layout is what makes the information are legible, clear and attractive.

According to Jenifer Tidwell, “Page layout is the art of manipulating the user’s attention on a page to convey meaning, sequences and points of interaction.”

In simple words, a page layout is the arrangement of various elements in such a way that a proper structure could be given to page. No matter how simple a page appears to be, but, behind the placement of even the minutest element like a dot or dash, a logic is there, provided the same is done professionally.

It takes a group of professional and skilled group of people to make an effective page layout. Several people work on the layout to make a graphic design. They include:

- Visualizer and Designer
- Photographer
- Illustrator
- Composer
- Finishing Artist
- DTP Operator

Different stages are taken care of by the above mentioned people. Designing a layout always involves research, thought and modelling. There are certain basic elements that are usually included in page layout, but there is really no limit to the innovations that can be adopted while deciding on the layout for a page. Thus, what needs to be taken care of is to keep a fair balance between the basic aesthetics and the scope of innovation.

### Check Your Progress 1

**Note:** 1) Use the space given provided below for your Answer.  
2) Compare your answer with those given at the end of this Unit.

1. What do you understand by page layout? Describe in your own words.

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2. Who are all the people involved in the task of page layout? Mention briefly.

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## 12.3 STAGES OF LAYOUT

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Layout is the arrangement of various elements like text and images on a two dimensional surface in order to make sure that all the information is legible, clear and attractive. For designing an effective layout, a designer must be very clear about the purpose of the design. Depending on the same, all the other aspects must be decided.

From idea generation to the final product, layout designing passes through various stages. Different degrees of finishing are associated with each stage.

**The basic stages of a layout are as follows:**

1. Visualization
2. Thumbnails
3. Rough layouts
4. The comprehensive

Let us now proceed to understand each of the above mentioned stages in detail.

### 1. Visualization

An idea first appears in the mind. After a long thought process, the same gets translated into a structured form. Then at this stage, the idea or concept thought by a communicator or designer can be translated into visual form by using a pencil or pen.

Also, during the visualization stage, the concept or idea is put onto the paper from the mind. It is the first stage among the various stages of layout planning. The more accurate and realistic the visualization is, the better the idea would turn out to be.

### 2. Thumbnails

This is the stage that primarily focuses on the visual part of the page. At this stage, different sketches are prepared by placing elements differently. This is the stage of trial and testing. During this stage, a very rough approximate size and shape of the element is drawn. Body copy or the text can be shown as square or rectangles.

**Advantages:**

- It can be done quickly
- It is economical (least use of colours, paper etc.)
- At least 6 trial layouts can be developed on one A/4 paper because of the small size

- Each trial provides new ideas
- Judging the ideas becomes easy

Although the above mentioned are some of the advantages of this stage, you, at the same time, it is very essential to draw the thumbnail sketches to the proportion. Thus, at every step, a balance of creativity as well as realistic approach is a must.

### 3. Rough Layout

Once the best option from the selected thumbnails sketch has been chosen, the same has to be drawn to the actual size of the finished layout. Here, all the elements are indicated clearly and accurately.

The rough layouts are generally used to give presentation to the editorial learn or very senior people in the organization. They are quite economical and effective while at the same time help to make the idea clear to the editorial learn.

The degree of finishing of the rough layout depends on the client's demand, time, amount of information with the designer and the skills of the designer as well.

In case of the rough layout, there are different levels of finishing ranging from very rough to well-finished.

### 4. Types of Layouts

In general, layouts can be classified into two types:

- **Working Rough** – It serves simply as rapid guide for the DTP operator, photographer and the printer who work in the background. The designer himself acts as the client and approves or rejects the working rough before preparing the finished rough.
- **Finished Rough** – The client many a times is not interested in the rough layout, but only wants to see the finished rough, i.e. the actual size of the finished product / design. All elements should be presented accurately with regard to size, style, spacing, placement etc. So, the final rough looks like a finished layout or something like a copy of original art will be pasted in place of the rough.

Earlier, all the rough layouts were prepared manually by skilled artists. But nowadays, with the use of computers and the related software, the rough layouts can be created in very less time without much effort and with more precision.

Rough layouts are mostly presented to the client in the form of a low-resolution printout or a soft copy in a CD or pen drive.

### The Comprehensive

This is the final step of the layout planning. It involves the making of a dummy layout of the page to be produced, showing the exact placement of page elements like text, graphics, colour etc. like a printed sheet. This will show how the layout will look in print. Once the draft has been approved by the designer or the concerned team, the same is also sent to the newsroom for final verification of the page design.

Illustrations or bromides are used on the layout and in case of digital comprehensives, software like CorelDraw, Illustrator or Photoshop are used along with a scanner.

The above mentioned stages are all inter related and the impact of one would make an impact on the next stage. Thus, it is very essential to ensure that each step is being carefully being executed.

### Check Your Progress 2

**Note:** 1) Use the space given provided below for your Answer.  
2) Compare your answer with those given at the end of this Unit.

1. What are the different stages of layout planning?

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2. How are the various stages of layout planning inter related? Explain in your own words.

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## 12.4 ELEMENTS OF LAYOUT

Irrespective of the idea or the budget there are always some fixed elements that must be present in order to make any layout visually appealing and pleasant to the eyes. Following are some of the basic elements of a layout:

### 1. Colour

Colour is a very important element in a layout planning. A poor choice of colour may ruin the whole layout. While choosing a colour, the theme, the idea behind and the target audience must be kept in mind. While talking of colour, both text and background colour must be taken into account. The text colour must contrast well with the background colour. It is advisable not to put light-coloured text on a light background or put dark-coloured text on a dark background as it will badly affect the ability of the reader to read the text.

Once a colour is chosen, the purpose of using it must be justified and a strong reason for it to be given. Colours are not present merely to make the design look bright and attractive. Rather, each colour has its own associations in the human mind. All such aspects must be taken into consideration while choosing any colour. In the next unit, you will learn more about colour in detail.

### 2. Text

When we talk of a layout, it is a combination of both – the text and the images. Thus, both the aspects need to be taken care of by the designer in order to assure

an effective result. As far as the text is concerned, it includes all the written part of a layout. Be it the body copy, the taglines, address etc. all are included in the text.

There are several attributes that are added to text like bold, italic and underline. Also, different fonts are available to choose from. With the help of a computer, we can create dramatic effects in the text very easily. But, what needs to be taken care of is the fact that text must be easy on the eye, clear and legible to the reader. The content must never be over powered by the effects.

### **3. Purpose**

The purpose of the layout must be absolutely clear to the designer. What is the layout all about? For whom is it intended? etc., all such questions must be answered before the task of layout is started. The purpose is what makes it clear as to what type of the final output is desired. For example, a layout of a women's magazine would be very different from the layout of a kids' magazine. Thus, it is the purpose that makes the work focussed.

### **4. Size**

The size of the page is another important factor that needs to be kept in mind. It is because knowing the page size tells us the exact space to work with. The size of a design must be in harmony with the page size to ensure effectiveness of the same. Therefore, it is always essential to know the page size before the process of planning for layout has begun.

### **5. Bleeds and Margins**

Most page layouts allow for bleeds in the design. As a designer, you should plan accordingly. A bleed basically extends the graphical elements past the document size so that when the piece is being trimmed, there are no white lines around the document. For that reason, you should also give yourself an appropriate margin in your document set up for important elements such as body copy or company information. Hence, keeping in mind the margin space and the bleed area is a must while planning any layout.

The above mentioned are some of the elements that need to be taken care of while designing any layout. As a designer, the aesthetic value must not be compromised for the sake of innovation.

### **Check Your Progress 3**

**Note:** 1) Use the space given provided below for your Answer.  
2) Compare your answer with those given at the end of this Unit.

1. What are the different elements of a layout?

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2. How, according to you, does the colour affect a layout?

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## 12.5 IMPORTANCE OF A GOOD LAYOUT

No matter how a layout planning is started, but, the aim of the designer is to arrange the different elements in such a way that they appear visually appealing and soothing to the eyes. Since there is no end to innovation, hence, there are numerous types of possible arrangements and patterns that can be thought of by the designer.

Layout plays a significant role in page designing. Proper layout can enhance the look of the page, thus, making the content stand out. The effectiveness of a layout depends on the arrangement of its individual elements as well as the overall visual hierarchy formed by them.

For a good layout, it is essential to keep the following elements in mind:

- 1) The arrangement should be such that the most important part of the content, be it in text form or the picture, must be located at a position which can grab the most attention. i.e. the dominant spot on the entire page. Usually, it is the top left corner of a page.
- 2) Colour scheme of the page should be in harmony with the content.
- 3) There should be some white space in the page to give a visual relief to the eyes.
- 4) The size of text should depend on the importance of the content. For example, the most important information is in the headline, so size of the same must be maximum and so on.
- 5) Despite being independent elements of a layout, the image and the text part must complement each other.

Apart from the above mentioned check list, it is very important that the layout of a page should be in accordance with the topic and keeping in mind the convenience of the readers/viewers. Otherwise, the reader's interest may be lost. This is where the role of a good layout comes into the focus. A good layout must be able to support the content to showcase it in the best possible way.

### Check Your Progress 4

- Note:** 1) Use the space given provided below for your Answer.  
2) Compare your answer with those given at the end of this Unit.

1. What according to you is a good layout?

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2. What are the essentials to be kept in mind while planning a layout?

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## 12.6 LAYOUT FORMATS FOR VARIOUS PRINT PUBLICATIONS

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All types of print publications are made with the help of a collection of several pages. Depending on the type of publication, the page layout is altered.

Let us now proceed to know the layout pattern for some of the common print publications.

### 1. Newspaper

Most newspapers are printed in two formats or sizes and they are:

- 1) Broad sheet (13"x 21")
- 2) Tabloid (11"x14")

#### Broadsheet

It is usually in the size 13 inches wide and 21 inches in length. The pages in a broadsheet are divided into 7-8 columns and each column is about 12 picas or two inches wide. The columns are separated by a gutter space between them. And as the content in a broadsheet is of serious nature, hence, clarity of the content forms the core focus rather than the elements of designing and the effects.

#### Tabloid

It is usually 11 inches wide and 14 inches in length. The pages of a tabloid usually contain 4-5 columns. The nature of content is considerably less serious in a tabloid as compared to broadsheet and therefore, there is ample scope for innovative layouts and placement.

### 2. Brochure

Brochure is primarily an advertising material. It may be used to advertise locations, events, products or services. The primary aim of a brochure is to give information to the reader which can be captured at a glance. Hence, the language used in the brochure must be simple and easy to understand. Also, the design should be eye - catching.

Brochures may come in all shapes and sizes and serve many purposes. It may even be produced from a single sheet of letter sized paper with two, three or four folds. Though a variety of formats are available for brochure design, the most commonly used format is 8.5 x11 inches vertical sheet with two folds, three folds, single fold and 11x17 inch horizontal sheet with three folds.

Keeping in mind the aim of brochure, the layout should be such that the text written in the same should be straight to the point and maximum information should be given in the form of bullet points.

The colours should be catchy and the text should be in a very conversational manner. The reader must not get the feeling as if he/she is being instructed to do something.

### 3. Pamphlet

A pamphlet is an unbound booklet (that is, without any hard cover or binding). It generally consist of a single sheet of paper that is printed on both sides and folded in half, thirds or fours to form a small booklet.

Pamphlets are cheap and easy to produce and are mainly printed for mass distribution. The quality of the paper used is also of low quality. A pamphlet can contain anything from information on kitchen appliance to medical information and religious treatises. Pamphlets have also been used as an important tool of political protest and political campaigning.

The layout for a pamphlet should focus more on the content rather than the designing. The text should be less and the message should be crisp and direct. There is less use of pictures in pamphlets.

### 4. Booklets

It is small book that carries a number of pages joined together, either by stapling or stitching. Booklets come in a variety of shapes and sizes but the most preferred size for the same is 4 x 8 inches.

Since, a booklet usually has more than two pages, thus there is scope of arranging the text as well as visuals on the pages. Also, if possible, distinct colour schemes can be used for every page. But, here also, the simplicity of language and avoiding the jargons (difficult words) is what needs to be specially taken care of.

### Check Your Progress 5

- Note:** 1) Use the space given provided below for your Answer.  
2) Compare your answer with those given at the end of this Unit.

1. What should be kept in mind while planning the layout for a tabloid?

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2. How is the layout of a pamphlet different from that of a booklet?

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3. What are the essentials to be considered while planning the layout for a brochure?

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## 12.7 LET US SUM UP

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In this Unit, we began with understanding a page layout. After getting to know about the page layout with the help of definitions and a detailed description of the same, various stages of the layout planning process has been discussed.

This was followed by the foundation bricks i.e. the basic elements to be considered during layout planning which have been explained in detail. Then, the importance of a good layout was discussed. Also, a checklist for a good and effective layout has been given. Lastly, the layout format for different print publications have been discussed.

Thus, an attempt has been made in this Unit to explain the page layout process beginning from the scratch and going step by step while bringing to light different aspects of the same.

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## 12.8 KEYWORDS

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|                      |                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Layout</b>        | : Layout refers to the arrangement of various elements on a page. It usually refers to a specific placement of image, text and style.                                                                                                                                                                                                                                       |
| <b>Design</b>        | : Design is that area of human experience, skill and knowledge which is concerned with the man's ability to mould his environment to suit his material and spiritual needs thus adding an essence of uniqueness.                                                                                                                                                            |
| <b>DTP Operators</b> | : A DTP artist is a desktop publishing worker, a job description used in advertising agencies, publishing, colour separation, printing and other related industries. A DTP operator is usually skilled in multiple computer design applications. DTP operators were formerly known as FA artists (FA: Finished Artwork). The name got changed due to the digital processes. |
| <b>Bleed</b>         | : Bleed is a printing term that refers to printing that goes beyond the edge of the sheet before trimming.                                                                                                                                                                                                                                                                  |

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## 12.9 FURTHER READINGS

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- 1) Tidwell, Jenifer, (2010). "Designing Interfaces". O'Reilly Media.
- 2) Klanten, Robert et al (2018). Newspaper Design. Gestalten.
- 3) Gautier, Ranien and Gaufier, Classic (2017). Design, Hypography, etc. : A Handbook, Niggli.

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## 12.10 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

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### Check Your Progress 1

- 1) A page layout is the arrangement of different elements – particularly the text and the visuals on a page in such a manner that all the information is clear

and legible.

- 2) Visualiser, Designer, Photographer, Illustrator, Composer, Finishing Artist and DTP Operator

### **Check Your Progress 2**

- 1) Visualization, thumbnails, rough layout and the comprehensive layouts are the main stages of a layout planning.
- 2) It is so because the work done in one stage would make an impact on the next stage. The impact of one stage – be it good or bad would influence the subsequent stage.

### **Check Your Progress : 3**

- 1) Colour, text, purpose, size bleed and margins are some of the elements of a layout.
- 2) Colour plays a key role in any layout. A poor choice of colour may ruin the entire impact of a layout. Also, each colour has certain associations in the human mind. Thus, while choosing any colour, the purpose of the same must be very clear in the mind of the designer.

### **Check Your Progress 4**

- 1) Layout plays a significant role in page designing. Proper layout can enhance the look of the page, thus, making the content stand out. The effectiveness of any layout depends on the arrangement of its individual elements as well as the overall visual hierarchy formed by them.
- 2) Importance of the content, colour scheme, white space, size of text, and harmony in the text and visuals are some of the essentials to be considered for a good layout.

### **Check Your Progress 5**

- 1) First of all, it must be taken into account that the nature of the content of a tabloid is less serious, thus, there is more scope for the use of innovative text styles, colours and the layout pattern. Thus, the flexibility of the planning must be there while planning a layout for a tabloid.
- 2) A pamphlet is mostly a single sheet of paper for mass distribution. Also, here, the quality of paper is low and the distribution is done to the people who are not the focussed audience. Because of the same, the content should be short, simple and straight forward. More bullet points should be used.

A booklet is a number of pages joined together and the audience of the same is relatively known to the client. Thus, there is more scope for the elaboration of text and also the use of images. Innovation in terms of layout pattern can be done in case of the booklet.

- 3) A brochure is a multi - fold advertising material. Keeping in mind the aim of brochure, the layout should be such that the text written in it should be straight to the point and more and more information should be given in form of bullet points.

The colours should be catchy and the text should be in a very conversational manner. The reader must not get the feeling as if he/she is being instructed to do something.

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## UNIT 13 PRINCIPLES OF PAGE DESIGN

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### *Structure*

- 13.0 Introduction
- 13.1 Learning Outcomes
- 13.2 Understanding Design Concepts
  - 13.2.1 Brief History of Graphic Design
- 13.3 Stages of Design Thinking Process
- 13.4 Elements of Graphic Design
  - 13.4.1 Basic Design Elements
- 13.5 Design Principles
  - 13.5.1 Design Process
  - 13.5.2 Harmony
  - 13.5.3 Rhythm/Movement
  - 13.5.4 Contrast
  - 13.5.5 Balance
  - 13.5.6 Unity
  - 13.5.7 Variety
  - 13.5.8 Emphasis
- 13.6 Relevance of Designing
- 13.7 Let Us Sum Up
- 13.8 Key Words
- 13.9 References and Further Readings
- 13.10 Check Your Progress: Possible Answers

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### 13.0 INTRODUCTION

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Everything in the world that surrounds us has a fixed pattern on the basis of which our eyes see the object or thing. The pattern which guides our vision is called as the design. Design is omnipresent and that is the reason that we always tend to look for a definition of the same. Design as a process is what drives the whole business cultures and makes sure that the environment surrounding us is easier to navigate. The single word ‘design’ encompasses numerous aspects, and that is why the search for a single definition leads to lengthy debate.

As every design is distinct from the other, in the same way, there are several definitions of design. Each definition has its own focus point and thus, none can be judged to be wrong or right. After understanding the very basic of design, let us now proceed to explain it in different dimension in depth.

## 13.1 LEARNING OUTCOMES

After going through this Unit, you will be able to:

- describe the basics of designing and designing as a process;
- use the concept of designing in your day-to-day life and above all for educational purposes; and
- apply designing in the teaching-learning process.

## 13.2 UNDERSTANDING DESIGN CONCEPTS

Graphic design is an umbrella term which encompasses the idea of page design. It is a way of arranging a message content creatively in order to attract the attention of the readers to any page. It may look like a simple process, but there are many intricate details which are complex to understand. As Archer said, that design is an area of skill, experience as well as knowledge that is related to the human ability to change his/her environment to match his/her spiritual and materialistic needs. Graphic design is well known as communication design. Cezzar (2016) states that design is “the art and practice of planning projecting ideas and experiences with visual and textual content”. Graphic designing is an integral part of almost all product advertisements, digital content, signage, etc. Within the packaging of digital content there are mainly *two* types of graphic designing- **interactive design** and **motion graphics**. Graphic designing is a part of applied arts. Design is a process to translate a construct into a blueprint. The translation of any idea directs the process of designing. Before inventing technology, making products or creating goods, the basic idea is always developed through the process of designing.

Be it a web designer, fashion designer, content designer, automobile designer or page designer, they are all form part of the graphic designing *genre*. Even though the main outputs may vary, but there are professional design styles which any designer must follow. Also, there are certain design principles that a designer must keep in mind while going through the process. The appropriate use of graphic and artistic elements is at the core of any kind of graphic designing

We must have understood by now that ‘graphic designing’ is essentially a creative process. In addition, it has also to deal with various social structures, cultural perspectives, aesthetics, political dimensions, economical conditions etc. in addition to design principles.

Each and every detail of a graphic design has its impact on the final output of the design process. It can range from the choice of colour, placement of various elements and shapes in a page or positioning and proportion of it indicating the complexity in the details.

### 13.2.1 Brief History of Graphic Design

The Altamira cave paintings in Spain are known for its charcoal drawings as well as the poly-chrome paintings of local fauna and human hands. It dates back from 34,000 to 15,000 BC? and they can be cited as examples of the human effort to express ideas in a concrete design form in the pre-historic era. Similar is the

case of the Lascaux in France, which is known for the Palaeolithic cave paintings which also dates back to 15,000 BC?

These cave paintings were used to communicate ideas, events and facts among people. There was a superimposition of ideas and thoughts, a different interpretation of the views of the painter's manifested idea or design. With the evolution of human civilization there were evident changes in the human thought process, lifestyle, etc. and consequently there are changes in the images that depicted human conditions. This was also affected by the change and advancement in technology.

Graphic designing is the creation of images and related elements. There are other segments of arts like applied arts, fine arts, etc. Designers or the artists use this medium as the means of communication for expression of their imagination. And it is known that the art of graphic design is usually done with a purpose.

With the technological advancement from pigment to pixel, the meaning of images also changed for human beings. The interpretation of images varies with each generation bringing changes in empirical thought in the process of interpretation. But there is a common message which is majorly shared by all the images created by the graphic designers.

The images would not work with the audience if proper research is not done to create the right impression for any type of information. Whereas there is a possibility that some information provided might not be interpreted properly by the audience due to some cultural constraints and their lack of experience.

### **Check Your Progress 1**

**Note:** 1) Use the space given provided below for your Answer.  
2) Compare your answer with those given at the end of this Unit.

1. What do you understand by a design? Describe in your own words.

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2. Is harmony an essential element of design? Give reason.

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### 13.3 STAGES OF DESIGN-THINKING PROCESS

As mentioned earlier, design process is a simple process but equally if looked at from a creative perspective it is highly complex in nature. Generally, there is no firm rules and regulations in a creative process, but the design thinking process goes through certain stages prior to finalisation into a design package. Here we shall discuss George Kembel's *five* stages of design thinking process: *Empathize*, *Define*, *Ideate*, *Prototype* and *Test*.

- a) **Empathize** - In this initial stage, the design team mainly devotes efforts to understand the newspaper readers' needs, behavior and thoughts about the news presentations. With the help of circulation and marketing departments, the design team interacts with the readers and observes their views. Basically at this stage, the design team collects feedback about the current news presentation package if it is the case of an already running newspaper but if it is the case of a proposed newspaper, then the design team collects the feedback about the readers' views on newspaper designs in general.
- b) **Define** - Once the feedback about the newspaper readers has been collected, the data will be collated at this stage. Collated data arranged in a systematic manner then goes through rigorous data analysis. Eventually, the results of this data analysis guides the design team to frame the objectives and goals for the new design package.
- c) **Ideate**- Based on the design objectives and goals, many possible new ideas are generated in order to meet the readers' expectations from a newspaper. Diverse combination of design elements are combined through working out different permutations and combinations in order to develop new design ideas. Besides the design team, other departments of the newspapers (management, editorial, circulation, and advertisement) are regularly consulted in order to seek their inputs for the new design package.
- d) **Prototype** - After diverse design ideas have been developed in the previous stage, the design team in consultation with the newspaper management and editorial team, shortlists a few design ideas and these selected ideas are further developed into full-fledged design packages for the newspaper.
- e) **Test**- A full-fledged prototype design package will be deployed for certain period for selective readers. It will be tested rigorously across different segments of readers in order to understand and identify any pitfalls and shortcomings. These testing trials helps to finalize the design package for the final deployment over the period of time

#### Check Your Progress 2

- Note:** 1) Use the space given provided below for your Answer.  
2) Compare your answer with those given at the end of this Unit.

Questions: Fill in the given blanks:

1. There are ..... phases of a designing process.
2. .... is the phase in which the designer looks for a fresh way to approach the designing.

3. Brainstorming, visual management, development methods and testing are the key activities of the ..... phase.

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## 13.4 ELEMENTS OF GRAPHIC DESIGN

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The elements of graphic designing are depicted in various ways based on both the focus as well as the objectives of the design output. Contextually, the concept of graphic design depends on the significance of the fundamental art elements. The design principles along with elements of the art is at the core of the designing process. Composition, typography, pictography and logo are the crucial elements of a page design. The outcomes of this process are the user interface design, web design and communication design.

### 13.4.1 Basic Design Elements

Exploring basic elements of design is an important stage in this process. In order to create an effective impression on the readers through your creativity and style, you should focus on the basic elements of design- line, shape, colour, texture, value and space. The arrangements of these elements in a proper manner is required to make the page attractive.

#### 13.4.2 Lines

Lines are the continuous marks on a surface from one point to another. They exist visually also. The quality of the lines depends on its thickness or thinness. Sometimes it may also have irregular shapes.

- (a) **Types of Lines** – You know there are a numerous types of lines around us. The qualities of the lines depend on various characteristic such as thickness, wavy, thin, straight and bold. From the borderline and the ones creating segments, these are all made up of lines.

Often lines create scenarios. The quality of lines provides meaning in the way it is interpreted by the receiver. Various types of lines depict different meanings according to the context. The imagination of the designer as well as the creativity used to depict news items are fundamental elements of the end product newspaper page. Hence, a designer should explore different possibilities to create logos using various types of lines.

- (b) **Functions of Lines** - Lines can be used in various ways. They depict grids, edges, borders and structures. Lines also make moods, impressions and expressions. Dynamic lines are formed through spontaneous brush strokes by designer. The intention and use of lines by the designer is an elementary part in the depiction of lines on any image. You can create interesting lines using the media (colour, ink, pencil, etc.) properly via the perfect medium (brush, charcoal etc.).

Lines are sometimes used to create different shades and varied tones. Forms, structures as well as the sense of shapes are defined by variable quality of the lines. If lines are stressed a bit, then a sense of tension can be created. Lines can also help in highlighting certain areas.

There are *two* purposes for which lines are often used they attract the attention of the readers to specific subjects. They can also make a difference between the two segments. In order to attract the attention of the readers

towards a section, thick and bold lines are used. For differentiation of segments thinner and dotted lines are used.

In the case of newspapers, however, you will note that predominantly bold lines are used especially for infographics as it highlights some important points which attracts the attention of the reader. In newspapers, you will also notice that thick and dotted lines are used to differentiate between columns of the news.

### 13.4.3 Shapes

Shapes can be either regular or irregular, depending on the confining area. They are separated into *two* groups - geometric and organic. The hard edged border shapes are the geometric ones, whereas the one with smooth border are the organic ones.

**Functions of Shapes** – Shapes provide, meaning to your design. The basic functions of various shapes are

- A. Meaningful Shapes** – Shapes create purpose to category and help to focus. It helps in highlighting and determining ideas in a specific way. It defines regions and provide space giving conscious importance to the news content. The mood and expression is defined by the shape of the news.
- B. Positive and Negative Shapes** - Negative shape is recognized in white and the positive shape in black colour. Shapes exist in space; negative shapes occupy positive space. Positive and negative space are also called figures and ground or foreground and background. Positive and negative shapes are often defined within their functions to describe space. Negative space is the space around and between the subjects of an image. The reversible effect is called the figure ground reversal.

An object which has the same focus point is not a negative space. As mentioned earlier, a negative shape is often referred to as a background to grab attention to the main subject, which is, the positive space.

Shapes have to be well-organized in space to create a good composition. The organisation of shapes in space should be well-balanced between its positive and negative shapes, however, it all depends on the intention of the designer. Shapes are sometimes organised to create a sense of expression or as an attention grabber.

### 13.4.4 Texture

You can view various kinds of texture around you. When you can feel a texture by touch as well as by viewing, then it is known as visual texture. The function of texture is to give meaning and to create visual function in graphic design. It is divided into *two* groups - tactile and visual texture.

**Tactile Texture** – When you can feel a texture by touching it is called Tactile Texture, for example- some surfaces are coarse while some are fine in texture. This is used in order to avoid friction between any two surfaces. It may also be used to improve the visual appearance of the whole environment.

**Visual Texture** - The texture which is produced as mark on any surface are Visual Texture. It can be created by brushing, rubbing, scratching, etc. The marks are viewed through light and colours.

Texture works as a visual aid is done before any of our designing activities. This is basically useful at the time of creating a composition. For example, if we combine bold lines with the fine texture it will depict the idea of calmness.

### **13.4.5 Form**

As we know that “form and shape imply space” it means that there is a requirement of space for any form to exist which can be produced by light and shadows. Forms are categorized into two groups- two dimensional (2D) and three dimensional (3D). The 2D forms are seen in 180 degrees, which is made from the setup of planes, according to the width and height (like in drawing and paintings) while 3D forms are seen in 360 degrees (like the sculptures), which are tactile in nature and can be felt through touch.

Forms can also be illustrated as organic and geometric like the shapes. The organic forms have smooth and rounded borders and lines, while the geometric forms have hard edge borders and lines.

**Illusions and Forms** – We have varied perceptions of different shapes. We see things differently and then interpret it in an altogether different way. The perception of a particular pose or form depends on the angle which is highlighted in the image and the feature which is emphasised of the object under scan. There is a kind of illusion created by various forms. These are created using various types of lines, series of lines, shapes, colours and even combination of lines or shapes.

### **13.4.6 Space**

Shapes that overlap each other and block the audience from the viewing a section of the object behind it make a sense of space within the shapes. When shapes are placed higher within the horizontal line it appears as if at a farther distance, therefore, giving a sense of space. Actually, size of the shapes create a sense of space.

Shapes which are smaller seem as if they are further away from the audience. The shape of objects which are at distant will be less detailed than the object which is closer to the audience. As you may know warm and cool colours also influence the illusion. Darker the colour the closer it will appear to the viewer.

Hence, space is a critical element of any package in the design. In the case of newspapers, we can relate it with the term “white space”. Space does not stand for any empty space, but it is a background which is filled with white colour, but there are occasions when the space is used with different colours. The main use of space, however, is to alleviate the human eyes to maneuver it among the words in the newspaper sections. Space usually provides a visual break. And a sensible use of white space attracts the attention of readers to the text.

### **13.4.7 Colour**

This segment talks about the value of colour and its application in graphic design. The following section on the meaning of colour discusses the functions of colour as well as the creation of colour which will be useful to you in terms of design application.

**a) The Meaning of Colour**

Colour is personal and subjective in nature in terms of its role in graphic design. It also sometimes functions to arouse one's feelings. People use colour according to their personal experience, social mindset and cultural background.

**b) Basic Colour Theory**

Colour is viewed according to the hue, tone and saturation. Hue is a common name for colours which has a huge number of variations from light to dark shade, for example red and blue. The varied range of colour is known as the tone. Colour varies from the light to the dark shades. According to pigment separation and in order to get a light colour you have to mix the white colour with some other colour such as red or yellow while on the other hand for a dark shade mix a colour with black. If you observe any diagram depicting the common colour mixing words, it will put tint as the mix of white that makes the colour lighter and shade is the mixing with black colour that reduces the lightness of the image colour. And tone is made by both tinting and shading. Some colour specialists are of the opinion that tone is also produced by mixing with grey colour.

The hue can vary according to the saturation or the chroma also. That is referred to as intensity. As you may be aware, saturation is the full intensity or something like proceeding from brightness to greyness. The temperature of a colour is based on its intensity. Lower intensity creates cold colours, whereas warmer ones are created by the higher intensity.

**c) The Colour Wheel - Pigment Mixture**

Birren (1965) stated that "in the mixture of average pigments, the primary colours are magenta red, yellow and turquoise blue. These combine to form orange, green and violet (or purple)". Therefore, secondary colours are made out of two primary colours.

**d) Additive and Subtractive Primaries**

Digitally, to predict how colour works, the colour system of Red, Green, Blue (RGB) which are also known as additive colours and the coloured pigments such as Cyan, Magenta, Yellow, Black (CMYK) which are also known as subtractive colours, never performs in a similar way. "If you are working with computer the colour on your monitor (RGB system) will not be the same as the colour that is printed (CMYK system)" (Dabner et. al., 2010).

**e) Contrast and Harmony**

Various types of colours support the content in a graphics. They give strength to the idea of a designer, which gets depicted by the core design. Therefore, a designer must understand the management of colour while applying contrast and harmony colours in design. The function and the perception of design largely depends on the choice and amount of colour. Colour sometimes changes according to the lighting effects. For example the same colour appears different when seen under normal light and sunlight. There are many factors that have a huge impact on how we perceive colours. It is a popular notion that you can achieve a good combination of colour and can also achieve clarity in the image if you use opposite colours in the foreground and background.

**f) Simultaneous Contrast**

When you arrange colours side by side or if some colours are embedded within it, different intensities are created. This concept is known as simultaneous contrast. On the basis of surrounding colours the central image is affected. Colours seem to be warmer if the colours surrounding them are darker. “Simultaneous contrast identified by Michel Eugène Chevreul refers to the manner in which the colours of two different objects affect each other. The effect is more noticeable when shared between objects of complementary colour” (Wikipedia, 2016).

**Check Your Progress 3**

**Note:** 1) Use the space given provided below for your Answer.  
2) Compare your answer with those given at the end of this Unit.

1. What are the different elements of design?

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2. What do you understand by texture?

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3. How do different shapes affect the human mind?

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**13.5 DESIGN PRINCIPLES**

To depict the imagination into graphics, a designer must follow some principles which are nothing but the organization of certain basic elements of design. The designer must understand each and every element in detail before executing the ideas onto a page. For any type of creation of a graphic designing product, it is crucial to understand the core elements of graphic designing.

**13.5.1 Design Process**

Visual research and references are the primary processes of graphic design principles. The point of reference researched by the designers works as the main point of the design process. When a designer explores various visual elements and applies them to their imaginative screen with all the creative skills, the process of design is initiated.

Designers must put in all their creativity and thinking ability along with their imagination power into use with the core elements of graphic design.

## 13.5.2 Harmony

Harmony is a term used in graphic designing to describe the synchronization of various elements of design - the organization of various components like shape and colour, contrast in colour, size etc. There are various elements which are interpreted in design and the organized combinations of these is called harmony in design.

## 13.5.3 Rhythm/Movement

A sense of visual movement is created into a design through the arrangement of visual elements, their movement or rhythm.

## 13.5.4 Contrast

As the name suggests, contrast is the stark differences between basic elements of graphic design in the composition. The pattern of contrast in the design relies on the characteristic differences between one or more basic elements of design.

## 13.5.5 Balance

Balance in visual effects is a sense created by the use of core elements of design that depicts contrasting elements as equally weighted and which shall not look odd. Both the sides of the composition are made in such a way that there is a balance through the equivalent arrangement within it.

There are many examples of a diverse pattern in balance – e.g. the shapes of the same size with a dissimilar arrangement but which is visually balanced. Unlike the size of shapes that are put properly in an equal manner still depicting balance one can also set shapes with dissimilar size in the middle of a composition, depicting balance.

### a) Symmetrical Balance

Symmetrical balance is similar to the original that too, around the central axis. The axis can be in any direction.

### b) Asymmetrical Balance

Asymmetrical balance is similar to the original, but it has different weight just around the central axis.

### c) Radial Balance

Radial balance is when the graphic elements are set based upon on the central axis as its focal point.

### d) Mosaic Balance

Mosaic balance happens in a chaotic balance. In most cases this is the result of a chaotic organization of core art elements and the lack of focal points wherein the vital art elements gets the same emphasis.

## 13.5.6 Unity

In the visual effects, the term unity is defined as the relationship between various visual elements which bear all other elements working as a whole in a combination.

The element of unity is when the perfect combination of shapes and colours that represent visual elements occur in *sync* with each other.

### 13.5.7 Variety

Variety works as a supplement of unity. It is crucial to create visual enjoyment. If a designer avoids variety it will lead up to monotony in the structure and the composition. It will create a sense of boredom. Unity and variety rely on each other in order to produce a good as well as an interesting design. The ability to use various art elements in multiple ways and in interesting combinations brings in the variety to a design. The designer can play with lines, their qualities, various shapes and sizes, hue, colour, etc to break the monotony of a composition.

Repetition of shapes and lines creates a sense of variety in a design which gets strengthened by the element of unity.

### 13.5.8 Emphasis

As the name suggests emphasis is to highlight a particular element of a design in order to attract the attention of the viewer to a particular element, structure, design or colour. This creates a central point to the whole composition and forms the most dominant area of the whole composition.

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## 13.6 RELEVANCE OF DESIGNING

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It is important to keep in mind that the very essence of any design is its attribute of uniqueness. Thus, in continuation with the same, the relevance of design also varies from industry to industry and from field to field. The following text would throw some light on this aspect.

### a) Design and the User

Good design begins with the needs of a end user. No design, no matter how beautiful and unique it may be, cannot be good if it does not fulfill the needs of its intended user. This may sound obvious but many products and services have failed because the people for whom it was intended could not connect with them. For example, milk – maid, which today is very widely used as a mithai maker was initially introduced in the Indian market as packaged milk, and it failed miserably. But, later the same product was introduced in the market as a sweetener. Today, milk - maid as a brand does not need any introduction as it is popular among the Indian masses as a sweetener.

Finding out what the customer wants is the first stage of what designers do. The designer then builds upon the results of that inquiry with a mixture of creativity and commercial insight. Although the initial trigger to start creating any design stems from the personal instinct of the designer, but apart from this, there are more scientific ways of making sure the design hits the mark. Different designers use different methods - combining market research, user testing, prototyping and trend analysis. And finally any product launch is ultimately a gamble which cannot be exactly predicated as to where it would reach, but these tested methods do help to decrease the risk of failure, a fact that often comes as a surprise to clients.

### b) Creativity

A design does not have to be new, different or impressive to be successful in the marketplace, as long as it fulfills a need, but design methods do lead to innovative products and services. The more impactful a design appears at the first glance,



the better it is expected to do in the future. Designers learn that ideas that may seem strange are worth exploring and that the ‘common-sense’ solution is not always the right one. Designers often hit on the concepts that they think need to be introduced through methods such as drawing, prototyping, brainstorming and user testing.

Close observation of specific habits of users in real-world situations gives insights into their behavior patterns that may lead to ideas that would not have been formed had the designer simply thought about the situation in isolation or relied on generalized market research. Hence, it would not be wrong to say that any design would not be an impactful one if it lacks a creative vision in the first place. Also, it must be made amply clear that even though creativity may be a quality acquired by birth but it needs to be definitely fine tuned with continuous practice over a period of time for successful application.

### c) Design and Business

Designers, unlike artists, can not be allowed to simply follow their creative impulses and their gut instincts. They work in a commercial environment which means there are many other considerations which have a bearing on the design process. Ultimately, the aim of every design, be it for a small scale or a large scale production is to turn the creation into profit. Designers have to ask themselves some questions before finalizing their designs such as: Is the product they are creating really in demand? How is it different from those already available in the market? Does it fulfill the needs of the intended end users? Is it viable? Will it cost too much to the manufacturer? Is it safe? and so on.

Companies have often designed their way out of a failure by creating a product that serves the customer’s needs better than its rivals through promoting innovative designs. However, as the sophistication of the consumer’s need and the competition in the global market increases, the uniqueness of the design becomes more and more valuable. In such a scenario, innovation in the form of design is the key to success.

### Check Your Progress 4

- Note:** 1) Use the space given provided below for your Answer.  
2) Compare your answer with those given at the end of this Unit.

1. What are the different principles of design?

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2. How does creativity affect design?

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3. Explain the relation between design and business.

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### 13.7 LET US SUM UP

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In this Unit, we have discussed about the basics of designing, to start with. In the introduction, the focus has been to explain the omnipresent element of designing.

After that, the design as a process has been defined. First, the general perspective about designing has been explained. Then, with the help of the quoted definitions, the same has been explained. Following the definitions, the distinction between designing and other activities in various industries have been explained. This was followed by the five major phases integral to every design process.

In the following text, the foundation bricks of design, i.e. the elements have been explained to make it clear as to how a final design is carved out using the combination of various elements like line, shape, space, colour etc.

Also, in the Unit, the principles that govern the process of designing have been explained followed by the explanation of how designing is relevant to different fields and walks of modern life.

Thus, an attempt has been made in this Unit to cover the designing process from all angles comprehensively beginning from scratch and through a step- by -step process bringing to shed light on different aspects of the designing process.

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### 13.8 KEY WORDS

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|                  |                                                                                                                                                                                                                 |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Design</b>    | : Design is that area of human experience, skill and knowledge which is concerned with the man's ability to mould his environment to suit his material and spiritual needs thus adding an essence of uniqueness |
| <b>Space</b>     | : It is every area of the design which is only filled with the background colour and not the other elements of design.                                                                                          |
| <b>Texture</b>   | : Texture is the feel generated through appearance or the physical touch of any surface, in general sense.                                                                                                      |
| <b>Dominance</b> | : Dominance means giving more prominence to any particular object in comparison to the other visual elements present in the design.                                                                             |

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### 13.9 REFERENCES AND FURTHER READINGS

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- 1) Malamed, Connie (2011). Visnal language for Resigners. Rockport Publishers.
- 2) Lupton, Ellen and Phillips, C. Jonnifer (2008). Graphic Design: The New Basics. Princeton Press.
- 3) Lidwell, William. (2010). 'Universal Principles of Design'. Rockport Publishers.

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## 13.10 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

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### Check Your Progress 1

- 1) Design is the creation of a plan for the construction of an object or a system which follows a unique pattern. It is what lays the foundation of the formation of any new object, system, idea, product or even service.
- 2) Yes, because it is the harmony which gives the essence of oneness in any design and makes it to appear like a single unit.

### Check Your Progress 2

- 1) Four
- 2) Discover
- 3) Develop

### Check Your Progress 3

- 1) Line, shape, texture, colour, value, space, etc. are some of the elements of a design.
- 2) Texture is the feel generated through appearance or the physical touch of any surface, in a general sense. Different textures give rise to different sense of feelings. The choice of textures in a design depends on the need of the same.
- 3) Different shapes give different sense of movements to our mind. For example: circle gives the sense of a continuous movement while square gives the sense of basic structures which have defined boundaries.

### Check Your Progress 4

- 1) Balance, gradation, unity, contrast, dominance etc. are some of the principles of the design.
- 2) For, all designs, creativity is a must. It is an inevitable part of every design. It is so because it is the creativity that gives a unique identity to every design.
- 3) Companies often carve their way towards the success by creating a product that serves the customer's needs better than its rivals in form of innovative designs. As the sophistication of the consumer and global competition in the market increases, the uniqueness of the design becomes more and more valuable.

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## UNIT 14 TYPOGRAPHY AND IMAGE

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### Structure

- 14.0 Introduction
- 14.1 Learning Outcomes
- 14.2 Typography
  - 14.2.1 Evolution and Brief History
  - 14.2.2 Principles of Typography
  - 14.2.3 Basic Elements of Typography
- 14.3 Typography Techniques
  - 14.3.1 Role of Typography in Design
  - 14.3.2 Creating Effective Typography
  - 14.3.3 Typography in the Digital World
- 14.4 Images
  - 14.4.1 Basic Functions of Images
  - 14.4.2 Types of Images in Graphics Design
  - 14.4.3 Importance of Images in Design
- 14.5 Images for Digital Media
  - 14.5.1 Using Images for Web
  - 14.5.2 Digital Images: Editing and Optimization
  - 14.5.3 Saving Images
- 14.6 Let Us Sum Up
- 14.7 Further Readings
- 14.8 Check Your Progress: Possible Answers

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### 14.0 INTRODUCTION

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In ancient times, people used to share ideas with others through talking and believed in oral culture. The messages were passed on from generation to generation by word of mouth for hundreds of years. The earliest writing systems known to man were pictographic. It means writing through pictures. The first people to use pictographs were the Sumerians, who live in Iraq about 5000 years ago (Sarkar, 2009).

The invention of typography by Gutenberg in 1447 revolutionized the printing technology to a great extent. Even though the method of block printing was invented more than 600 years before Gutenberg by the Chinese monks, it was he who embraced the technique of moveable type and solved the problem of slow speed. He simplified the process of movable type which enabled the arrangement of the types in any order as per the requirement of the printing.

Images play a significant role in media production. Today, visual images can be seen everywhere, whether in print, electronic media or even in new media. In fact, visual images complement the text that is being communicated and help to attract the attention of the public. We are all quite familiar with the saying- “a picture

tells a thousand words” which aptly fits the real value of the images. Images enable us to learn effectively, grab attention, enhancing the communication concepts and inspire.

In this Unit, we shall discuss the evolution of typography and its various techniques as well as role of typography in design. We shall also discuss the basic functions of visual images and their role in the digital media.

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## 14.1 LEARNING OUTCOMES

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After working through this Unit, you will be able to:

- describe the term typography, its growth and development;
- discuss the principles and basic elements of typography;
- discuss various typography techniques;
- define the basic functions of images; and its types; and
- undertake editing and optimization of digital images.

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## 14.2 TYPOGRAPHY

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Derived from the Greek word *typos* means “to form” or “impression” and *graphein* means “to write”, typography is the art and technique of arranging types (letters) in linear form to make the written language readable when displayed on paper. Even after the advent of online media, typography holds the same importance when displayed on “electronic screen” as well.

### 14.2.1 Evolution and Brief History

One of the most fundamental forms of human communication is writing. Pictographs were considered as the oldest system of writing. As the name suggests, pictograph means writing through pictures. It is believed that the Sumerians had first started communication through pictures and developed a writing system whose wedge-shaped (cuneiform writing) strokes would influence the style of scripts. In cuneiform writing, figures were pressed onto soft blocks of clays giving them a unique wedge shape. Cuneiform has gained universal acceptance in the Middle East despite the fact that it was not a language, but a set of symbols. This form of writing was passed on from one culture to another.

Hieroglyphs (means carving) was the Egyptian way of writing which was generally made up of pictures of different natural creatures like animals, plants, birds and the other things. To represent ideas in front of the world these pictograms were used by ancient civilizations, and they soon evolved into alphabets leading to the development of various typographic systems.

Movable type was first invented in China which generally involved ceramic or wooden components. In medieval Europe during 13th century, typographers used single letter tiles to create letters on the walls. This form was very popular in Europe until the invention of metal movable type by Johannes Gutenberg in the 15th century. His technique allowed letter punches to create multiple copies of the same document, and the first movable type book “the Gutenberg Bible” was printed.

### 14.2.2 Principles of Typography

As a human beings, we observe only things minutely which are placed and synchronized with our surroundings. And only if things are placed in a proper manner do we usually appreciate them. The same thing works for print media, web media and its allied areas like selecting page layout, graphics design and other forms of art which are being in 'sync' with the end product. In the context of typography if it is visually looking good, people get attracted to it. Hence, for people who primarily work in the design industry, it is very important to know the rules of good graphics design and typography so that they can create elegant work. Since there is no limitation to creativity, designers can use different options in typography, but there are some basic principles which should be taken care of while creating any graphic.

There are two major aspects of typography: legibility (or readability) and aesthetics. Tracy (1986) described legibility as the quality of being decipherable and recognizable. It means that the typographers should select the correct fonts relevant to the content which is being produced. Every content has some specific target audience. For example if you are designing a children's magazine, you can use comic sans font for the text. On the other hand if you are designing a newspaper, you may select serif font (for e.g. Times New Roman). Readability refers to how easy it is to read the text as a whole. The correct use of document margins, kerning and leading and correct layout impacts the readability. Tracy (1986) opines that some fonts or font styles, for instance sans-serif fonts, are considered to have low readability, and so be unsuited for large quantities of prose.

The aesthetics in typography basically deals with the entire document (for print and web). It is based on the selection of typefaces (fonts) and their different sizes, layout of the page and colour decisions. Some of the common rules are listed below to achieve the best results:

Correct decisions for types and for the layout should be relevant to the content.

- a) Use of two typefaces on a page only if they can be harmonized with each other.
- b) The leading (line space) should be pleasing to the eyes.
- c) Headlines should not be placed at the bottom of the page.
- d) Text should be placed in a manner so as to lead the eye from one text to another in a natural way.
- e) There should be no use of ALL CAPS in the headlines, unless it really justifies them. ALL CAPS can be used in kicker headline.

### 14.2.3 Basic Elements of Typography

The basic elements of typography are the following:

#### Consistency

One the most important principles of typography is consistency. In any typographical work, the consistency in using typefaces, spacing, leading and formatting is a very important aspect. It lets you to make your work more professional and help your readers to concentrate on the content rather than formatting of paragraphs and lines.

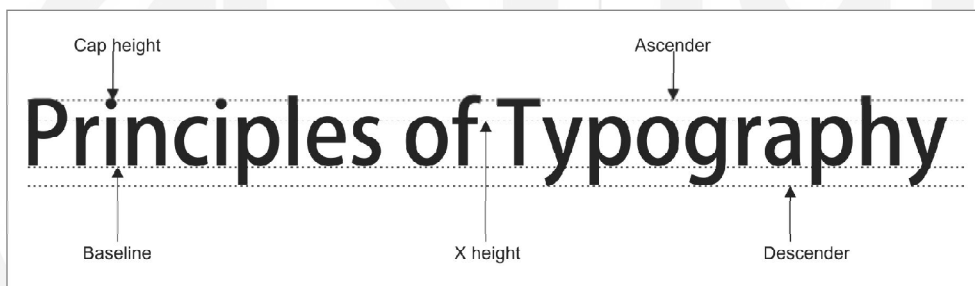
In any graphic design whether you are designing a book, a newspaper or a magazine or even a web page, the hierarchy in placing text in appropriate order intentionally helps your reader to go through the content in faultless sequence. For example, in a book, the readers' eyes first reads the title, then author's name and after that the chapter and its content. The prime objective of the hierarchy in typography is to follow the sequence (text) in order to prevent any distraction for the readers.

### Alignment

Alignment helps to unify the appearance of a text piece. The text alignment mainly places the cursor or aligns the text with the document's different margins. There are four types of text alignment – a) *right alignment*: starts each line in the document at the right most margin; b) *left alignment*: it is the default alignment in the Microsoft Word in which each line starts from left; c) *centre alignment*: starts each line in the document from the centre; and d) *justified alignment*: aligns text with left and right margins horizontally on page margins.

### Lines

Any text line has five sub lines which are aligned with each other as mentioned in Figure 1.



**Fig. 1: Functions of Lines**

- a) *Baseline*: Where the text actually sits on. One of the basic positions where the text actually sits. In primary level schooling, this is one of the first lessons children learn.
- b) *Cap height (or cap line)*: Marks top position of the capital letters.
- c) *Ascender*: Some letters extend above the x-height. Here *i*, *l*, *f*, *t* and *h* are the ascenders.
- d) *x-height (or midline)*: Describes the height of lowercase letters excluding ascenders and descenders. Typically measured as the height of the letter *x*.
- e) *Descender*: Some strokes drop below the baseline, Here *p*, *y* and *g* are the descenders.

### Leading

It denotes the space between two lines. Leading is obtained by measuring distance (in points) between two baselines. Decreasing leading gets lines closer and make the text paragraph "tighten". On the other hand, when you increase the leading, the paragraphs consume more white space on page. See the example below in Figure 2:

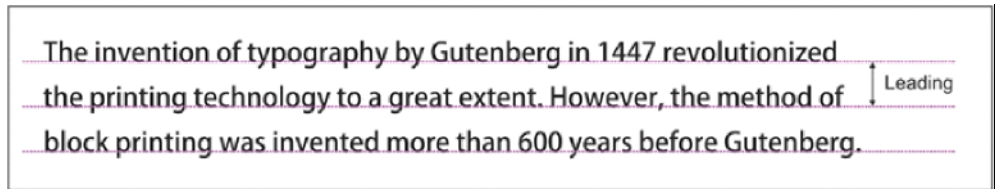


Fig. 2: Leading

## Tracking and Kerning

In typography, tracking means the space between groups of characters. It is also called letter-spacing which is based on space between words in one text block. Tracking or letter-spacing may be described as loose or tight. Loose tracking is when the letters have a greater distance, when the letters are closer it can be describe as tight tracking.

Kerning describes how much space there is between two characters. Tracking and kerning are often confused with each other. Tracking has a global setting which suggests the distance between all the characters. Kerning is basically a microscopic view of the space between two alphabets. See the example below in Figure 3.

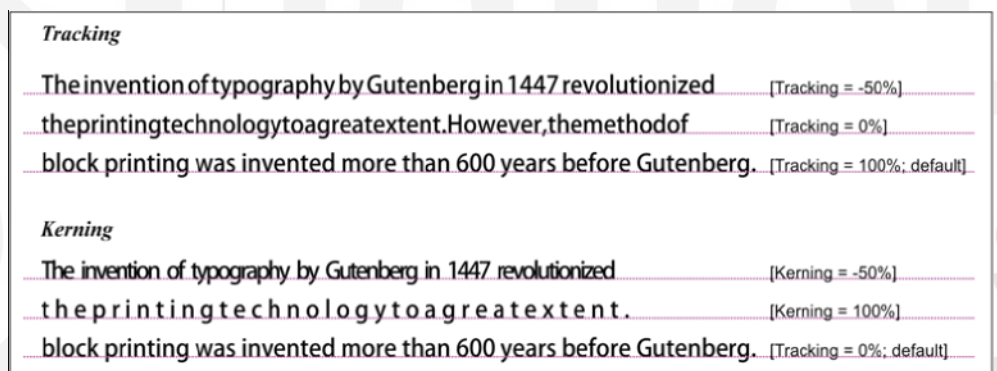


Fig. 3: Tracking and Kerning

## Check Your Progress 1

**Note:** 1) Use the space given provided below for your Answer.  
2) Compare your answer with those given at the end of this Unit.

1. Define typography in one or two sentences.

.....

.....

.....

.....

2. List three necessary elements of typography.

.....

.....

.....



3. Differentiate between tracking and kerning.

.....

.....

.....

.....

## 14.3 TYPOGRAPHY TECHNIQUES

There are many techniques of typography which can be employed in graphics design. It has many facets and different techniques e.g. spacing and kerning, specific spacing for specific size, x-height and vertical proportions, variations in characters, width, weight and contrast etc. and several techniques that must be taken into account when considering the suitability of space. These techniques are used for organizational strategies and attractive qualities when placing two or more different and/or contrasting fonts together. For example you can use simple sans-serif fonts for headline, they will help the titles in grabbing more attention to the story.

### Typefaces

What do you understand by typeface and a font? Are they the same? Let us clarify the basic thing here. As discussed above in this Unit, typography is an art of creating letters, designing them in a way that they look real in the document (whether for print or web).

Typeface is a set of characters with common features of design. Some types also have different font weights and styles—Regular, Italic Regular, Bold, Bold Italic. A font, on the other hand, is a letter collection or combination. Today, thousands of fonts are available. The fonts are essentially a typography mechanism for presenting your message through documents. The distinction between these two was described by Brownlee (2014). According to him, Garamond would be the typeface: he described all the thousands of metal blocks that a printer might have on hand, designed with the same basic principles of design. But a font was something else entirely. A font described a subset of blocks in that very typeface—but each font embodied a particular size and weight. For example, bolded Garamond in 12 point was considered a different font than normal Garamond in 8 point, and italicised Times New Roman at 24 point would be considered a different font than italicised Times New Roman at 28 point.

Now with the internet we have a much wider variety of old and new types to read and use. These types provide us a number of options for our design productions. There were limitations earlier so one had to choose from a few available typefaces but now there is vast variety of types.

### 14.3.1 Role of Typography in Design

Phinney (n.d.) opines that typography is like fashion, or furniture. With rare functional exceptions, the world doesn't "need" new clothing or furniture designs, but people want to look different or evoke a particular feeling or fit with a particular "look," and there are trends and styles.

In graphics design, typography can have a major impact on how people react to your document. Typefaces should be selected carefully and consistently. The appropriate use of graphics, colours and images is equally important to create a professional brand. Selecting and using the right font lets you to induce a particular sentiment or certain style to your audience. The skillful use of typefaces grabs the attention of your audience and communicates the key idea of the document so that they can act on it. Any choice made by a graphic designer, including fonts, has an effect. Yadav (2014) opines that there is a science or a scientific purpose behind different functions of typography for example, kerning, leading, columns, point size, line length, etc. It is much more than just letters and words on a page. In much the same way that tone of voice can alter a message, typography has power to communicate a great deal simply through the forms of the letters and their layout on a page.

Bernard (2019) compared eight popular typefaces to find which one performed the best and at what size. The fonts that were compared were Century Schoolbook, Courier New, Georgia, Times New Roman, Arial, Comic Sans, Tahoma and Verdana. These fonts were compared at 10, 12 and 14 point sizes. Results were as follows: At 10 point size, Verdana was the most preferred font. At 12 point size, Arial was the most preferred font. At 14 point size, Comic Sans was the most preferred font.

For the effective use of typography, one has to anticipate appropriate use of fonts and styling which affects your audience consciously or sub consciously. The skillful use of typography can affect the emotions and performance of readers in other tasks.

Levi (2008) has indicated that visual fatigue from reading off a computer screen can be ascribed to 'awkward character rendering, poor choice of line length and other typographical elements'. This highlights the importance of good typography to help online readers reduce eye fatigue and eye strain.

### 14.3.2 Typography in the Digital World

The expressive use of typography has many facets, and many different techniques come with them to help with visual aid and graphic design. Spacing and kerning, size - specific spacing, x - height and vertical proportions, variation of character, width, weight, and contrast are several techniques that need to be considered while considering or creating the appropriateness of specific types.

**Colour Contrast:** Colour contrast may seem like a basic concept, but while selecting colors for your typography and overall design, there are some easily overlooked things to note.

**Font Size:** 10 pt font is generally accepted for body copy in print design. But on the web 10 pt is 13 px equivalent and this is a good size to stick with on the web for body copy. Anything smaller than these sizes is going to be too small to be read by the average viewer.

**Leading:** Leading is the space between text lines – what web designers call it as line height. The word leading originated in printing presses when type was set by hand. Lead strips have been placed between type lines to add space.

**Hierarchy:** Web designers often use the tags < h1 >, < h2 >, and so on to establish typographical hierarchy. Hierarchy is not just about size, but more about the prominence of your typographic elements relative to each other.

**White space:** White space or negative space in a composition is the space between elements. If your typography and other elements of design are dense and too close together, it will be difficult to read your content. Whitespace can be used to create balance or to turn the viewer's eye from one part of the composition to another. It may invoke a feeling of elegance or add to a typographic treatment to a level of communication.

## Check Your Progress 2

- Note:** 1) Use the space provided below for your Answer.  
2) Compare your answer with those given at the end of this Unit.

Fill in the blanks.

1. ....means the space between two lines.
2. .... in a composition is the space between elements.
3. A .....is a set of characters with common design characteristics.
4. ....font is generally accepted for body copy in print design.
5. .... describes the height of lowercase letters and excluding ascenders and descenders.

## 14.4 IMAGES

Images help us learn, images attract attention, images explain and help us to follow tough concepts and images may even inspire. The human brain is capable of visual processing as is shown by our love for images. We usually tend to get attracted to them as soon as we see images and are immediately drawn towards them and they hence serve as good attention grabbers. So we find extensive use of visual images in newspapers, electronic media, magazines, posters and other display materials every day.

Images are processed at an alarming rate by the human mind. Also human beings are able to analyze data quickly when we see a picture, and come to know immediately the meaning and context within it. It takes only a fraction of milliseconds for the human brain to recognize a familiar object or face.

In the context of graphics design, Sarkar (2009) defines that the “visual images are important graphics and images that are used for visual communication which may be hand-drawn, painted, photographed, or generated digitally on the computer”. Graber (1996) has noted that “combining pictures with words makes the message more memorable”.

### 14.4.1 Basic Functions of Images

The primary function of images is to attract the reader's attention and provide additional information to the text. Images assist the plain text by doing the following functions - explain, instruct, and entertain.

So in any communication material, the images do play a key role. Images have a powerful impact on the attitude of a viewer which cannot be created by text alone. The presence of a photograph will significantly change the attitude of a reader towards the framed issue in a news context. To attract the attention of readers, newspapers which are largely textual use different graphics design elements, images, blurbs, infographics etc. which essentially include images. Many newspapers routinely use half-jackets which are usually full of images and graphics and blow-ups of incidents. The photographs in a news story serves to draw attention to a big story, further illustrate statements made in the story, and give more details about the presented information to the readers. In feature articles, the use of images is important as they attract attention of the readers.

Images also play a key role in advertisements. One cannot imagine any advertisement without an image. "Seeing is believing" - images increase credibility of a message. An image, a video, or a graph is not merely a way of getting the audience's attention, but it is also a way of winning their confidence. A few catchy phrases about your brand will make your target market look your way, but a nice picture or an interesting video of your product and service would ensure a positive response from your audience.

Multimedia graphics are one of the most important components of electronic and digital media. Multimedia is essentially an integration of text, graphics, drawings, still and moving images (video), animation, audio, and other such media where any kind of information can be digitally represented, stored, transmitted, and processed. Images in a multimedia play a very important role. It is expressed in the form of still image, painting or a digital camera photograph. The points at which an image is sampled are referred to as picture elements, commonly known as pixels.

In a nutshell, images are used in almost all the media forms – print media, electronic media, digital media, animations, social media and so on.

### 14.4.2 Types of Images in Graphics Design

To create images, graphic designers use various techniques of art. However, whether they work with photography, digital art programs, or old - fashioned media such as pen, ink or paints, they still focus on conveying ideas. Images can be divided into two important types - denotative and connotative images.

#### **Denotative images**

Images that simply display an object or thing of representation are denotative images. They are simple, straight forward, and have no deeper meaning than what you see. Denotative images can be made as a very life - like outline, or just as an outline. For example – picture of an apple or a picture of a man sitting on a beach.

#### **Connotative images**

These are still pictures, but they can provide the viewer with more information and a broader context. They are used for creating an atmosphere, telling a story, or making a deeper meaning and connection. For example – a picture of a man who is sitting alone on the beach: it can show the man's solitude, depending on the interpretation of the viewer and possibly their own personal experiences, could present an essence of sorrow, tranquility, or thoughtfulness.

### 14.4.3 Importance of Images in Design

Usually, images are the best way to get ideas across in a graphic design. Ever since photography has existed, it has been used as a key advertising component. The images used in design as graphics are more than simple images. Designers need to know how to create images that convey specific ideas or connect with consumers effectively. All images are intended for a specific purpose in a graphic design. Images can mean a great deal in aesthetic communication. As a visual communicator, photography is essential in a variety of ways for the graphic designer. If you have multiple images in a single layout, you need to keep searching for a way to work together or try to incorporate other elements. When dealing with the style of ‘art’ imagery, other elements such as symbols, typefaces, colours, boundaries, etc., become very important.

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## 14.5 IMAGES FOR DIGITAL MEDIA

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In every digital media platform, images play a major role in engaging users with the website content. Images certainly add context and divert the attention of the users to deeper engagement with different topics. To add context to the digital there are multiple tools present – photo galleries, video galleries, motion graphics, infographics etc. In simple words, one topic can be explained in multiple formats. Since there is no space constraint in digital media, the pictures are used almost with every topic, story or article.

### 14.5.1 Using Images for Web

Images have improved the online experience since the dawn of the internet and continues to be an integral part of website design. Images are the major visual element in a website and appears in logos, backgrounds, sliders, and much more. They draw users to read the text, illustrate articles and blog posts and sometimes just entertain. Contextualizing through images in websites is a great idea. In digital, there are multiple tools available - photo galleries, videos, infographics and other contextual links. The topics are explained in multiple formats so that people can get engaged with the content.

#### Preference of audience

A website without images is boring and most of us would probably prefer one with lots of pictures, even with a nice design. There are many reasons why we believe images to be attractive. Rather than reading a lot of text that takes time and effort, we scan an image or a graph rather quickly. Images can then trigger all kinds of emotions and memories, making them very engaging. Images can help to cut across language barriers much better than text.

#### Images trigger emotions and engagements

In this context, images play a major role to engage people with the content. Images are an easy way to enhance the website’s user experience. They provide users the deeper context and assist in deeper engagements with the website content.

#### Using meaningful images on the web

Use of images on the website should have some logic. There is no need to make webpages heavy by putting a number of pictures that are less relevant to the content because they convey a sense of support for your message.

### **Using real images**

Using human images on the website is an efficient way of engaging the visitors. It makes us feel like connecting with other people, not just some coded website.

### **Background images**

Background images create a unique website atmosphere or gives the visitors an immediate impression about the website.

### **Avoiding slowing down of the website**

While images can have a major impact on a website's user experience, simultaneously, they can also slow down the website and irritate users at times. Due to good internet connectivity and the increasing smartphone penetration in India, most of the people see the content through their hand-held devices. Optimizing images is important so that they can be easily viewed on smartphones. It is possible to optimize their size or use thumbnail images to preview larger ones.

### **Images on Social Media**

Using images in social media is a great way to catch viewer's attention as it performs better than text on social platforms. Using relevant images on social media can help you get attention, build loyalty, communicate vital information, and increase engagement.

## **14.5.2 Digital Images: Editing and Optimisation**

To save images optimised for the web, there are three main file types: GIF, JPG, and PNG. Each type of file has its own strengths and weaknesses, and knowing and keeping these in mind when saving an image is very important.

### **JPEG (Joint Photographic Experts Group)**

The most popular file type for images on the web is JPGs (also known as JPEGs). They are ideal for photographs or complex pictures with lots of colours, shadows and gradients. JPGs handle these kinds of images well because they have a vast palette of colours to work with. It is also possible to save JPGs in high quality, low quality or anywhere between them. This allows you to accurately adjust and save the image, balancing the quality and size of the file. JPEGs are the most popular file format online and are widely used for product images, photographs and homepage banners / images.

### **PNG (Portable Network Graphics)**

Another popular online file format is PNGs. You will have the option to save PNGs in Adobe Photoshop. Most importantly, transparency can be handled by PNGs. This is one of the biggest difference between PNG and JPEG. Here, transparency is meant by a picture without having any background which can be used in any print or web documents and matched with their original background. PNGs are generally used for logos because they are of high quality and can handle transparency. It works well for logos because logos almost always require transparency.

### **GIF (Graphics Interchange Format)**

GIFs were popular many years ago, but are still an option for small image sizes that require just a few colours. GIFs can be used for logos but are not popular in most cases of e-commerce due to their limited colours option.

### 14.5.3 Saving Images

It takes longer to load large images. Large means the file size rather than the image dimensions, i.e. the KB, MB, GB values. Research shows that 47 percent of users expect a web page to be loaded in less than 2 seconds, and 40 percent expect a page to be loaded in not more than 3 seconds. It is therefore important that the images are small enough to ensure a quick site. To optimize images for web the following steps can be taken:

**Right dimensions:** The images should be viewed 100% in Photoshop. This allows you to view the image exactly the size it will display on a computer monitor when you save the web image.

**Saving for web:** The file size can be reduced without significantly reducing the image quality. The popular way to reduce file size without decreasing the image quality much is to use the “Save for Web” function of Photoshop. Photoshop images allow you to choose the quality of the export. It drops down the file size to below a megabyte and also there is no noticeable quality difference.

**Images compression:** Image compression applications are the easiest way to reduce image file size. These tools provide a quick and easy way to reduce the size of files without losing the quality of images. ImageOptim, ShortPixel, Kraken, Image Optimizer are some of the applications which compress the size of images effectively.

#### Check Your Progress 3

**Note:** 1) Use the space provided below for your Answer.

2) Compare your answer with those given at the end of this Unit.

1. Describe two major types of images used in graphics design.

.....  
 .....  
 .....

2. What do you understand by multimedia graphics?

.....  
 .....  
 .....

3. What are the steps to be taken before saving or optimizing images for web?

.....  
 .....  
 .....

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## 14.6 LET US SUM UP

In this Unit, we have discussed about the role of typography and image. We have traced the growth and development of typography and its principles. We have discussed the basic elements of typography such as consistency, hierarchy, alignment, lines, leading, tracking and kerning. Various typography techniques have also been included for better understanding of the readers. We have looked at the role of typography in design. We have also discussed the role of typography in the digital

world and its expressive use in the digital platform by discussing the colour contrast, font size, leading, hierarchy, leading and whitespace.

While discussing the images, we have traced the basic functions of images in communication and media. The role of images in different media forms have been discussed in detail. The types of images in graphics design was also discussed in which we touched upon two most important types of images i.e. denotative and connotative images. In addition, the importance of images in design, images in digital media was also included. We have also discussed the use of images for web editing and optimization of images in digital media in detail.

To sum up, it can be said that in communication, typography and image play an important role. Even though typography has a long history it has evolved significantly over a period of time. The pictures play a key role in any communication material. Images have a powerful impact on the viewer's attitude, and this impact cannot be created by mere text alone.

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## 14.7 FURTHER READINGS

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Yadav, P.; Chakraarti,D; & Bisoyi, D. (2014). Typography as a statement of Design. Conference Paper. International Ergonomics Conference HWWE 2014.

Bernard, Michael & Lida, Bonnie & Riley, Shannon & Hackler, Telia & Janzen, Karen. (2019). A Comparison of Popular Online Fonts: Which Size and Type is Best?.

Phinney, T (n.d.). How to Explain Why Typography Matters. Retrieved from <https://www.commart.com/columns/how-to-explain-why-typography-matters>

Levi, M.D. (2008). Literature at the Human-Computer Seam. Retrieved from [https://www.bls.gov/osmr/htm\\_papers/st000100.htm](https://www.bls.gov/osmr/htm_papers/st000100.htm)

Graber, D. (1996). Say it with pictures. The Annals of the American Academy, 546, 85-96.

Sarkar, N. N. (2009). *Art and Print Production*. Delhi: Oxford University Press.

Tracy, W. (1986), *Letters of Credit*, Gordon Fraser.

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## 14.8 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

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### Check Your Progress: 1

1. Typography is the art and technique of arranging types (letters) in a linear form to make the written language readable when displayed on paper.
2. (i) **Consistency** : In any typographical work, the consistency in using typefaces, spacing, leading and formatting is a very important aspect.  
(ii) **Hierarchy**: In any graphic design whether you are designing a book, a newspaper or a magazine or even a web page, the hierarchy in placing text in appropriate order intentionally helps your reader to go through the content in faultless sequence.



(iii) **Alignment:** Alignment helps to unify the appearance of a text piece. The text alignment mainly places the cursor or aligns the text with the document's different margins.

3. **Tracking:** In typography, tracking means the space between groups of characters. It is also called letter-spacing which is based on space between words in one text block.

**Kerning:** It describes how much space there is between two characters.

### Check Your Progress 2

1. Leading
2. White space
3. Typeface
4. 10pt.
5. X-height (or midline)

### Check Your Progress 3

1. **Denotative images:** Images that simply display an object or thing of representation. They are straight forward, meaning no deeper than what you see. Denotative images can be made as a very life - like outline, or just as an outline. For example – picture of an apple or a picture of a man sitting on a beach.

**Connotative images:** These images are used for creating an atmosphere, telling a story, or making a deeper meaning connection. For example – a picture of a man who is sitting alone on the beach: it can show the man's solitude, depending on the interpretation of the viewer and possibly their own personal experiences, could present an essence of sorrow, tranquility, or thoughtfulness.

2. **Multimedia graphics** are one of the most important components of electronic and digital media. Multimedia is essentially an integration of text, graphics, drawings, still and moving images (video), animation, audio, and any other media where any kind of information can be digitally represented, stored, transmitted, and processed. Images in a multimedia play a very important role.

3. To optimize images for web, the following steps can be taken:

*Right dimensions:* The images should be viewed 100% in Photoshop. This allows you to view the image exactly the size it will display on a computer monitor when you save the web image.

*Saving for web:* The file size can be reduced without significantly reducing the image quality. The popular way to reduce file size without significantly decreasing the image quality is to use the "Save for Web" function of Photoshop. Photoshop images allow you to choose the quality of the export. It drops down to below a megabyte the file size and there is no noticeable quality difference.

*Images compression:* Image compression applications are the easiest way to reduce image file size. These tools provide a quick and easy way to reduce the size of files without losing the quality of images. ImageOptim, ShortPixel, Kraken, ImageOptimizer are some of the applications which compress the size of images effectively.

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## UNIT 15 INFOGRAPHICS

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### Structure

- 15.0 Introduction
- 15.1 Learning Outcomes
- 15.2 Understanding Infographics
  - 15.2.1 Historical Background of Infographics
- 15.3 Types of Infographics
  - 15.3.1 Informational Infographics
  - 15.3.2 Timeline Infographics
  - 15.3.3 Charts Infographics
  - 15.3.4 Pie Charts Infographics
  - 15.3.5 How to Infographics
  - 15.3.6 Process Infographics
  - 15.3.7 Comparison Infographics
  - 15.3.8 Number Infographics
  - 15.3.9 Resume Infographics
- 15.4 Creating Infographics
- 15.5 Visual Elements of Infographics
- 15.6 Tools of Infographics
- 15.7 Let Us Sum Up
- 15.8 Further Readings
- 15.9 Keywords
- 15.10 Check Your Progress: Possible Answers

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### 15.0 INTRODUCTION

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Information design is the practice of creating visual presentations that communicate information for the purpose of instruction. Infographics is an exciting visual way to tell a story, reveal the truth in the data, provide engaging and accessible information. If they are timely, relevant and useful – and great to look at - they can have a visual impact. But the content has to be rich, well-researched, accurate, and have a strong narrative flow.

In this Unit, you will learn how to find the story in the data and tell it to your readers using tried and tested journalistic techniques, from angle and structure to powerful headlines and an irresistible call to action.

You will explore different infographic formats and develop content all the way to the point of design. The Unit covers the power of infographics and why they are worth considering in your communication efforts. It lays the foundation for building great infographics.

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## 15.1 LEARNING OUTCOMES

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After reading the Unit, you will be able to:

- understand data visualisations process which leads to creation of charts and graphs; and
- describe the balance between words and graphics for an impactful outcome.

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## 15.2 UNDERSTANDING INFOGRAPHICS

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There is a popular saying, ‘a picture is equal to thousand words’, which has become amplified manyfold after the introduction of infographics in recent times within the news presentation in dailies and magazines. In a simplest definition, infographics can be described as a visual representation of news, data, and information through combination of various media elements. Infographics is a process of organizing various visual and textual elements to convey a complex idea in a simple way using text, photographs, and graphical elements colourfully.

There are two ways of representing data in a visual format - data visualization and infographics. Data visualization in a standard format in which data is presented in the form of charts and graphs. It does not provide much scope for interpretation of data and showing relationships between data in this type.

In the infographics, however, a pattern gets established and relationship of data also gets clearly identified. Yet there is no standard set of visual tools for infographics. Data visualization is part of infographics, but infographics is not part of data visualisation.

### 15.2.1 Historical Background of Infographics

The history of infographics is about 32,000 years old. If you pay attention to the cave paintings from 30,000 BC you will be convinced that they were in fact, the first infographics. They depicted animals, weather, and all the things that the painter saw around him.

Similarly, Egyptian hieroglyphics from approximately 3000 BC is also an example of infographics as they were used to tell stories of the days, nights, work, rulers, animals, prayers and Gods.

In the year 1786, extensive use of bar charts, pie charts, line graphs and histograms representing the economy of England was seen in the book named ‘The Commercial and Political Atlas’. This became a milestone in the modern history of infographics. This book authored by William Play, makes him an early innovator in the field of Statistical Graphics.

In 1857, the famous nurse from England, Florence Nightingale used infographics in order to leave a strong impact and move the heart of the queen. With the help of coxcomb chart, an amalgam of stacked bar and pie charts she changed history. She successfully persuaded Queen Victoria to work towards improving conditions in military hospitals during the war. Nightingale’s chart reflected the exact number and causes of deaths during every month of the Crimean War. She also used colours for clarity. For example, preventable diseases were marked blue, wounds were marked red, and other causes were marked black.

Not only this, the biggest step in the history of infographics was also taken in England. Back in 1933, Harry Beck created the first map of the London Tube which showed lines to depict transit routes for public and tube stations. This was a big step because this showed how the infographics could be used in daily life and was accepted as a useful tool due to the ease and clarity it offered to the people. It played an important role in easing out the hassle for the tourists and travelers. This encouraged the use of visual diagrams in everyday life.

Even the road signs we see today all around us are an example of infographics. In 1972, Otl Aicher created a set of pictograms to be used during the Munich Olympics. It featured stylised human figures which further inspired the generic road signs we use today. For example, the sign of a man crossing the road, sign for road under construction, and work in progress etc.

Further, in 1975, Edward Tufte along with John Tukey, conducted a seminar on Statistical Graphics. In 1982, Tufte, a pioneer in the field of infographics published Visual Display and established himself as an infographic expert. Hence, he is called the father of data visualization.

Coming to the recent times, history of infographics is vivid and more dynamic. The use of infographics has reached almost all spheres of life now. Office oriented softwares like Excel and Power point are one of the most easy to use software that enable integration of infographics. In academic fields like economics, statistics, research analysis etc. infographics plays a vital role.

When it comes to historical aspects of infographics from the journalism point of view, the USA Today takes credit for popularising visual news presentations in early 1980s. USA Today's infographics attracted significant attention from the readers and it helped this newspaper to achieve higher level of circulation. Subsequently, contemporary news media organizations such as the New York Times and Times magazines are using infographics extensively, particularly the Times magazine is known for its spectacular infographics.

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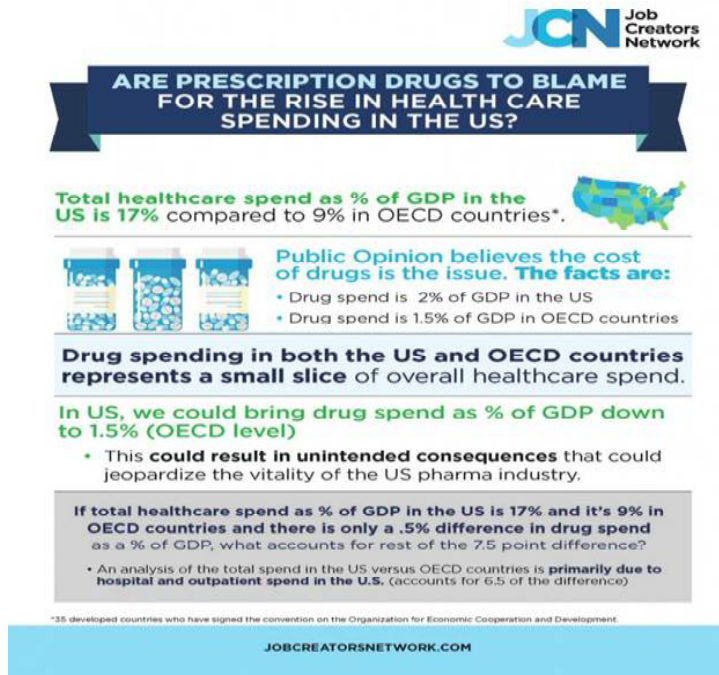
## **15.3 TYPES OF INFOGRAPHICS**

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If you have a trouble in choosing the type of infographics to use, understand your audience first. If your aim is to simply break values, or make a comparison, then choose the type of infographics that solves your purpose. Here are some commonly used types of infographics.

### **15.3.1 Informational Infographics**

Informational infographic is easily identified by the amount of words used in it. The priority of an informational graphic is to use more words in order to pass on enough information to the audience. The average use of words in this type of infographics is different from the other types and hence, you will be able to differentiate by noticing the number of words used. You will find the graphic is enhanced by colours, icons, shapes and other visual elements but overall, the emphasis is on the words.



via Job Creators Network

### 15.3.2 Timeline Infographics

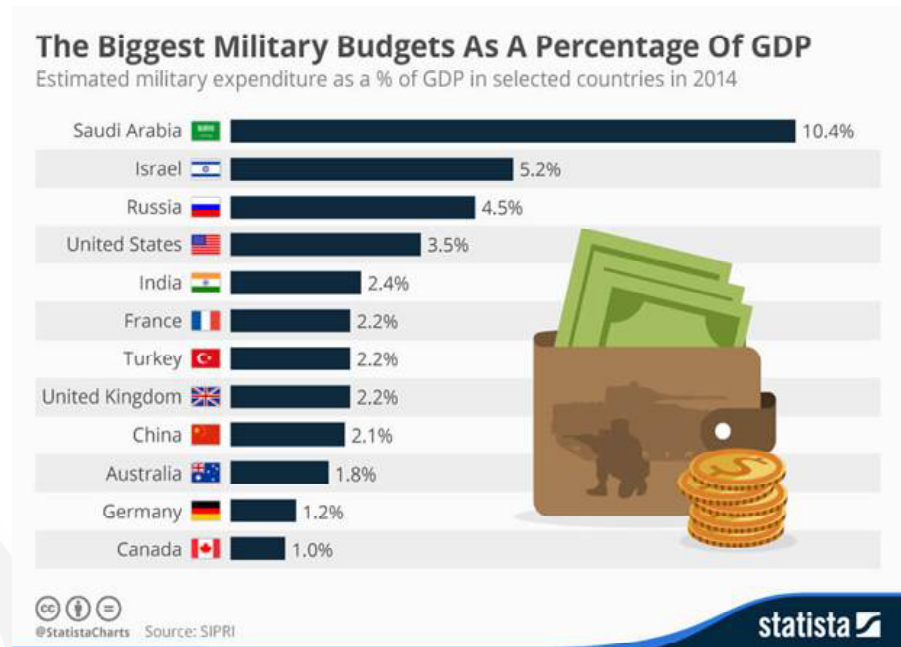
The word 'Timeline' is itself self explanatory. There is a chronological order which is used to depict the events or actions. They are mostly used to present an idea's evolution, a product's development or a historical trend. Timeline infographics includes icons, images, and graphic elements in order to put the point across. The timeline's format varies. It can be vertical, horizontal or winding but vertical and winding timeline infographics are comparatively easier to read and understand. The best utilization of horizontal timeline infographics is in posters, presentations, and environments where there is freedom and space is not a constraint.



via citifmonline

### 15.3.3 Charts Infographics

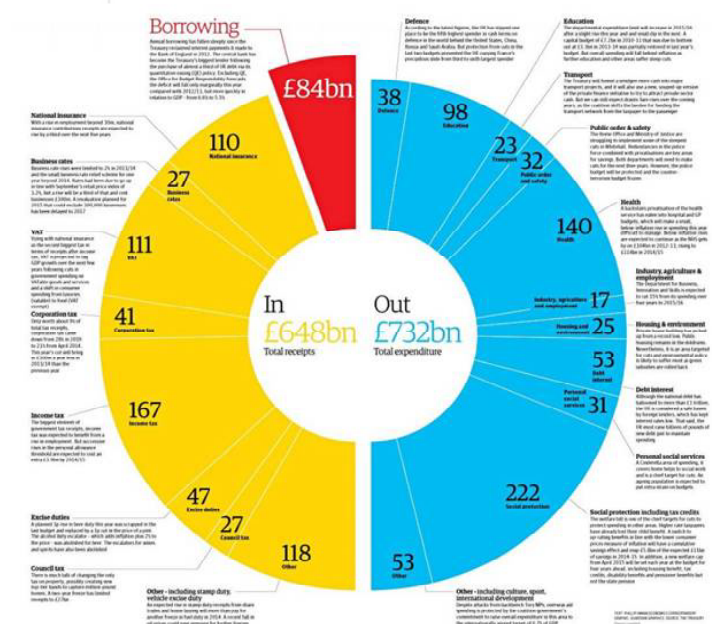
This type of infographics has a chart as the centerpiece of the information visualisation. Colours, shapes, and icons can also be added for a better impact, emphasis and elaborated explanation. This infographic is the best choice if you are performing a basic comparison of items. Chart infographics are the most relevant if you are comparing the number of students in various schools, Instagram followers of different actors and celebrities or the population in some chosen cities.



via Statista

### 15.3.4 Pie Charts Infographics

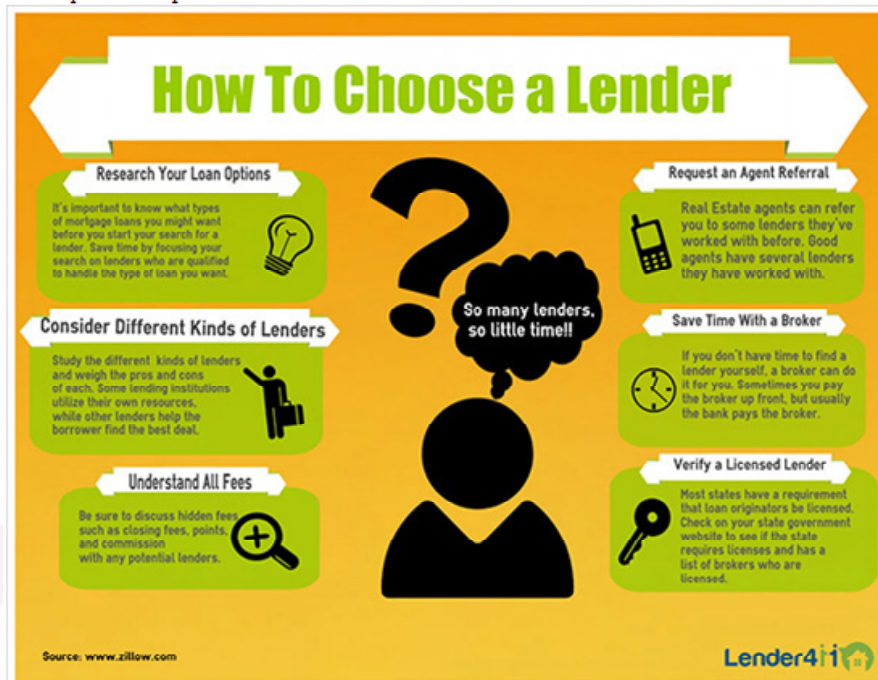
This infographic is actually a subset of a chart infographic, but we assign it a different category due its widespread use. In this type of infographic, the centre object is a pie chart. Pie charts are utilized best to depict different components within an item different and how one value is compared with other similar values.





### 15.3.5 How to Infographics

A systematic, step wise progression can be seen in this type of infographics. It has more of an instructive tone to its outlook and is best used for recipes or DIY creativity. How to infographics depicts the steps involved in achieving a certain outcome. It can be used in showing a product's manufacture, an ISO certification, or a medical procedure. The definition of this type of infographics in simple words is that each step is a logical consequence of previous one.



via Lender411

### 15.3.6 Process Infographics

This infographic is quite similar to a how to infographics. The prime distinction is that a process infographic presents decision-making processes. Process infographic is also called decision trees and is commonly known as flow charts. Every step is linked to the next one with lines or directional arrows which help in understanding the direction of the flow of information. Examples of situations where this type of infographics can be applied are personality type determinations, assembly line manufacturing, software programs, project plans and process audit.

We Make It Simple to Retrieve Tax Transcripts from IRS.  
**Learn How We Accomplish It.**



via 4506-Transcripts

### 15.3.7 Comparison Infographics

A comparison infographic helps in break-down of the similarities and differences between two or more things. These can be products, locations, events, actions, ideas or individuals that can be compared. Various situations where this type of infographic can work are: comparison of Managers vs. Leaders, in Sourcing vs. Outsourcing, or Past Strategy vs. Present Strategy. This type of infographic helps in creating a contrast and helps in identifying the pros and cons of one item with its alternative. You will see the graphic usually split into two or more sections which depends on the number of items that you wish to compare.





### 15.3.8 Number Infographics

The way information infographics emphasises on words, this type of infographics mainly emphasises on numbers. Situations where the numbers matter the most e.g war hit people, flood hit people, financial break up of a firm's investment etc. These examples and many more can be efficiently portrayed by this type of infographics.



via ISTE

### 15.3.9 Resume Infographics

Resume plays a vital role in applying for a job. The potential employer saves time and gets clarity about the work experience, professional experience, qualification, basic traits, and qualities etc of the applicant by looking at the resume. Here the resume infographics helps in imparting clarity by highlighting the key points and if made properly, it can leave a good first impression also. Hence, for an extra score this type of infographics can be used. The best use of this infographics can be seen while applying for a job or while promoting your services to the audience.



via Big Interview

## Check Your Progress 1

**Note:** 1) Use the space provided below for your Answer.  
2) Compare your answer with those given at the end of this Unit.

1. Differentiate between data visualization and infographics?

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2. Describe Timeline infographics?

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3. Compare 'Number' and 'Process' infographics?

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## 15.4 VISUAL ELEMENTS OF INFOGRAPHICS

The word infographics is a combination of information and graphics. Hence, the information which is being presented using various forms of graphical designs and patterns will definitely have a lot of visual stimulation. We are fascinated by colours and patterns around us as they are the perfect stimuli for our eyes. Hence, it is important to know about the elements of infographics in order to leave a long lasting impression on the audience.

### 15.4.1 Colours

Colours are the most eye-catching aspect of infographics. Hence, we will address this topic in detail. The colour scheme plays a vital role. It can make or break your infographics. Colours have the ability to either highlight or obscure information.

If you work on an infographics where you compare a country's major political formations, you can use red for political party no. 1 and blue for political party no. 2. This way you can make the graphic easy to understand.

What you need to understand here is that colours can be used very well to differentiate between different subjects that you want to compare. Differentiation is an important outcome. While comparing items, you can use colours to show a stark contrast between the two. For instance, black and grey would turn out to be poor choices to demonstrate how different two types of motor-bikes are.

It is important to consider colour of the text first while choosing the colour of the infographics. If you choose white text on a yellow background it will be a strenuous read. It is suggested that you choose colours based on the background colour of the page where you wish to ultimately place the infographics.

You might have seen an infographics with a white background on a website with a white background, it is not wrong but it can make it a bit difficult for readers to figure out from where the infographics begins and ends.

If you are working on infographics of a particular brand, it is a good idea to use the official brand colours as much as possible for consistency.

Some colours have the ability to evoke certain emotions and it is a good idea to take advantage of their ability. For example, blue and red colour are considered as colours of stability and trust respectively. Hence, they are used extensively in financial service brands infographics while the colour green goes well with environmental issues because its connotation.

The key factor is to keep things consistent. This has to be adhered to not only in the context of colours but all other elements as well. If you have used a particular colour scheme at the beginning of a series of infographics, it will be a good idea to stick to it in the following infographics on the same thing. This creates a recall value for the reader and it becomes easy for him to follow the story with time. Consistency is important even within the infographics. You should not show an infographics element in one colour in the beginning and change it later. Shifting of colour choices for the same element in the same infographics must be totally avoided.

### 15.4.2 Fonts

The number of options available for fonts has increased drastically in recent years. There are now more infographics fonts options as compared to the usual Times New Roman, Arial, Calibri and Comic Sans. Fonts can be classified in several ways but the four broad groupings by experts are Serif, Sans-Serif, Script, and Decorative.

**Serif:** it has small lines attached at the end of each letter and are more formal in nature. For example Merriweather and Times New Roman.

**Sans-Serif:** literal meaning ‘without serif’ and they do not have the small lines. They are modern and more streamlined than Serif fonts for example Arial, Helvetica and Lato.

The Script fonts give a handwriting feel and can be seen as fun, casual or elegant whereas decorative fonts have a peculiar form of novelty to them (e.g. the snowcapped fonts on Christmas graphics) They can be used sparingly and for specific subjects.

Comic Sans is a kind of font that is not ideal for a majority of business related infographics as it appears too casual. The fonts you choose will give readers the first impression of the content hence you should choose wisely.

Once you are done with establishing the purpose of your graphics, look through the options of infographics fonts available on your software’s editor’s library or you can download some more fonts from the internet.

Once you are done with choosing the type of font, you step on the next vital task. You will have to choose between light, bold or italic versions of the font.

On one hand, the bold fonts are good for headlines or for the sections of the infographics you wish to draw the most attention to. Headers and sub-headers must have a larger font than the rest.

One important thing to keep in mind while choosing fonts for the infographics is not to excessively dazzle the reader. You need a font that can help explain the content of the infographics on its own and not become a task in itself.

### 15.4.3 Icons

Icons are usually taken for granted but they can actually be of great help while placing a lot of information in small spaces. It is almost like a magic to see how a single icon can replace a paragraph.

The first step is to have an eye for items that can be replaced with an icon. For example, the word mango can be replaced with an icon, even running, sleeping walking can be shown using an icon.

You are a good graphic designer if you have the knack of replacing words with icons to make the infographics more effective.

### 15.4.4 Images

Images must be used in a way that they do not take over the infographics else you will convert the graphic into a patchwork of images with a smidgen of text. The advantage of using images is that they can replace acres of explanations.

Some tips for good choice of images are avoid clichés i.e stock photos, avoid repetition of images, etc. as the reader can sense repetition and lose interest easily. Even when the infographic is of an original idea, the presence of cliché images degrades its impact and changes an exciting message to a boring one.

## 15.5 CREATING INFOGRAPHICS

The main golden rule for creating infographics is that data should be presented in the multimedia mode. The purpose of infographics is to simplify a complex idea for easy understanding, at the same time to present it in a clear and accurate manner.

In infographics, text plays a central role along with other elements such as photographs, charts and diagrams. Fonts used for presenting data should be simple in nature and the text should be readable. Deliberate efforts should be made to avoid presenting the text in a decorative manner for the sake of attracting the reader's attention. Text in infographics may be various positions - be it headline, quotes, numbers etc. but relative and relevant font sizes should be maintained so that it will take care of the hierarchy of importance.

Similarly, a holistic colour harmony should be created within the infographics by avoiding the cluttering of the data with too many design schemes and colour codes. Also remember that use of many colours may become counter-productive, if a reader is visually impaired.

According to Sneh Roy (2009), infographics can be mainly divided into three parts, namely 1] Visual, 2] Content, and 3] Knowledge. Roy further classifies these three broad categories into further divisions. The following table shows the matrix of components of infographics:

| Visual          | Content      | Knowledge  |
|-----------------|--------------|------------|
| Colour Coding   | Times Frames | Facts      |
| Graphics        | Statistics   | Deductions |
| Reference Icons | References   |            |

Roy recommends *five* stages for creating infographics: 1] preparation of sketches and flowcharts - basically in this stage a blueprint is planned for the infographics; 2] devising a colour scheme - as per design principles, a minimal approach for using colours would be an effective way. Accordingly, a suitable colour scheme needs to be identified. 3] identification of graphics to represent the data or content. 4] a proper research needs to be undertaken to collect relevant data for the infographics. Authentic data should be collected from reliable sources. 5] With the help of graphics, colours and text, a relationship between these elements needs to be established. Through this process a new meaning can be conveyed to the readers. The ultimate aim of follow these *five* stages is to create a infographic that simplifies a complex data.

Hopkinsons (2011), on the other hand, outlines 10-steps to create an effective infographics:

Step 1: *Determine Timing* - decide required time available and fixing a deadline to create the infographics.

Step 2: *Provide direction*- identify your objectives and goals of your infographics.

Step 3: *Decide on payment* - finalizing required budget for creating the infographics.

Step 4: *Organize the data* - do your preliminary research to identify the required data.

Step 5: *First pass*- create your first draft of the infographics.

Step 6: *Getting to the “Eureka” moment* - discussing the draft infographics with the newsroom staff or other colleagues for better clarity and concurrence.

Step 7: *The first “final” candidate is sent* - after receiving inputs from the newsroom, prepare your final draft.

Step 8: *Make it better* - discuss your final draft of the infographics with the newsroom staff or other colleagues before giving the finishing touches.

Step 9: *The Final product* - infographics is finalized for publication.

Step 10: *Spread the word* - once it gets published, share it with the world through social media.

Few generic guidelines for creating infographics:

1. As a designer, your understanding of data for the infographics is crucial.
2. Your understanding makes infographics simpler for others to understand the complex data set.
3. Include details about the primary data source in your infographics works.
4. Do not clutter with too many content/topics. One topics covered in depth in a single infographics is an effective one.
5. Photographs, brand names, symbols and icons will convey the idea of infographics in a simpler way.

Now you have learnt ways in which an infographic can be created. Let us move on to know the available digital tools or software to create an effective infographics.

### Check Your Progress 2

**Note:** 1) Use the space provided below for your Answer.

2) Compare your answer with those given at the end of this Unit.

1. Name any two media elements needed to create infographics?

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2. Give one example each for serif and non-serif fonts?

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3. Name three components of an infographic?

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## 15.6 INFOGRAPHICS TOOLS

It is not just fancy images and catchy colours that makes an effective infographics. The design element is what makes it special. Choosing the right tool is the first step to enter the world of infographics. Some of the major tools are given below:

### Piktochart

Piktochart has over a hundred templates to choose from. You still have the liberty to create your own templates. The plus point of this tool is that new templates are added every week.

It is free software and you can also subscribe to it by paying around \$15 every month. Website address of Piktochart is <https://piktochart.com>

### Canva

You can avail this one for free or upgrade to 'Canva for Work' for \$12.95 per user/month. It is a popular online design tool and has a vast inventory of templates. This one has a user-friendly interface and is comparatively easy to use. Even those without a tech or design background can smoothly use it.

Some of the online websites which offers services to create infographics are given below. Many of these sites require registration while free subscription are available to anyone to create infographics.

1. amcharts visual editor (<https://www.amcharts.com/>) - online service that helps to create charts
2. Easil.ly (<https://www.easil.ly>) - helps to create various kinds of infographics
3. Pixlr (<https://pixlr.com>) - Online Photo Editor
4. Tagcloud (<https://tagcrowd.com>) - helps to create tag/word cloud
5. Infogram (<https://infogram.com>) - online visualization tool
6. Venngage (<https://venngage.com>) - free infographic maker
7. Gapminder (<https://www.gapminder.org>) - helps to create infographics on world issues



## 15.7 LET US SUM UP

In this Unit, we have discussed the historical background and the evolution of infographics in the mass media field. As you have learnt, infographics is rich in graphics and its main attractive feature is its ability to attract the reader's attention towards crucial facts and figures of an event or a big national story.

Infographics conveys information in a nutshell and effectively in a short time. It also occupies comparatively less space. In addition, you have learnt about the various types of infographics and its specific purposes. These types of infographics would give you an idea of the utility and purpose of infographics in the news field. Importantly, you have learnt about the various graphic elements involved in creating infographics, particularly the meaning and role of each media element in conveying news to the readers.

## 15.8 KEYWORDS

|                           |                                                                                                                                                                                                                                                                                  |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Infographics</b>       | : In other words it is called information graphics. Visual representation of a data set or a complex news content through charts, diagrams, photographs or a combination of all.                                                                                                 |
| <b>Data Visualization</b> | : In data visualization, graphical representation of data is the primary objective rather than creating a new knowledge or meaning out of data sets. Whereas in infographics, beyond representations, a relationship between data sets is established to generate a new meaning. |
| <b>Information Design</b> | : A branch of professional design discipline, where both infographics and data visualization falls under. Here in information design, it deals with the creation of data, categorization of data and visual presentation of data.                                                |

## 15.9 FURTHER READING

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## 15.10 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

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### Check Your Progress 1

1. There are two ways of representing data in a visual format - data visualization and infographics. Data visualization is a standard format in which data is presented in the form of charts and graphs with not much scope for interpretation of data as well as relationships between data existing in this type. In infographics, a pattern gets established and relationship between data also gets clearly identified. There is, however, no standard set of visual tools for infographics. Data visualization is part of infographics, but infographics is not part of data visualization.
2. The word 'Timeline' in itself is self explanatory. There is a chronological order which is used to depict the events or actions. They are mostly used to present an idea's evolution, a product's development or a historical trend. The timeline's format varies. It can be vertical, horizontal or winding but vertical and winding timeline infographics are comparatively easier to read and understand.
3. The way information infographics emphasises on words, number infographics mainly emphasises on numbers. Situations where the numbers matter the

most e.g war hit people, flood hit people, financial break up of a firm's investment etc. Process infographics is also called decision tree and is commonly known as flow chart. Every step is linked to the next one with lines or directional arrows which help in understanding the direction of the flow of information.

### **Check Your Progress 2**

1. (a) Icons are usually taken for granted but they can actually be of great help while placing a lot of information in small spaces. It is almost like a magic to see how a single icon can replace a paragraph. (b) Images must be used in a way that they do not take over the infographic else you will convert the graphic into a patchwork of images with a smidgen of text. The plus point of Images is that they can replace acres of explanations.
2. *Serif*: it has small lines attached at the end of each letter and are more formal in nature. For example Times New Roman.  
  
*Sans-Serif*: literal meaning 'without serif' and they do not have the small lines. They are modern and more streamlined than Serif fonts for example Arial, Helvetica and Lato.
3. According to Sneha Roy (2009), an infographic can be mainly divided into three parts, namely 1] Visual, 2] Content, and 3] Knowledge.